

Uterine and Tubo-ovarian Histology and Pathology

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MEDICINE**



Histology Resources

Michigan Histology Laboratory Manual [A link to an online histology lab guide](#)

Histology Guide <https://histopathology.pennstatehealth.net/slidebox/slidebox.html>

Digital Histology <https://digitalhistology.org/>

Histology Primer(American Board of Pathology) <https://histologyprimer.abpath.org/Modules/Index>

Shotgun Histology [A link to a series of Youtube videos](#)

University of Utah <https://webpath.med.utah.edu/HISTHTML/HISTO.html>

Duke University Histology <https://histology.oit.duke.edu/>

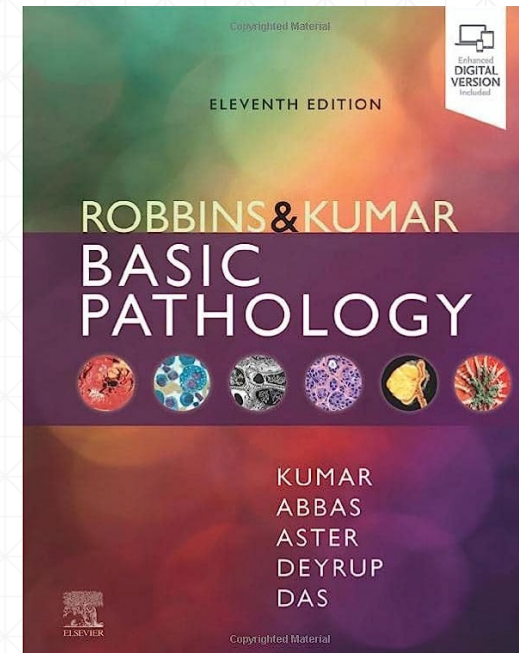
University of Leeds UK <https://www.histology.leeds.ac.uk/index.php>

Netter's Histology Textbook

Wheater's Histology Textbook

Pathology Resources

- PathElective <https://www.pathelective.com/>
- Kurt's Notes <https://schaberg.faculty.ucdavis.edu/>
- Interactive Pathology Lab <https://peir.path.uab.edu/wiki/IPLab>
- **WebPathology.com** <https://www.webpathology.com/index.asp#>
- **Robbins Pathology Textbook (11 th. Edition.)**



Objectives:

- Describe the microscopic organization of the ovary, uterine tube, and uterus
- Relate histological changes to the ovarian cycle, menstrual cycle, and pregnancy.
- Explain how altered structure/function leads to clinical consequences (e.g., follicular cysts, tubal infertility, endometriosis, abnormal uterine bleeding, menopause).
- **Recognize congenital and developmental anomalies of the uterus and their clinical implications (Self study).**
- Explain the pathophysiology and clinicopathologic features of functional disorders, including polycystic ovary syndrome, endometrial hyperplasia, adenomyosis, and endometriosis.
- Discuss pelvic inflammatory disease and infections of the uterine tubes and endometrium: common pathogens, pathogenesis, morphology, and complications.
- Compare benign and malignant uterine tumors (e.g., leiomyomas vs. leiomyosarcomas), including morphology, clinical course, and prognosis.
- Differentiate between type I and type II endometrial carcinomas in terms of risk factors, molecular basis, morphology, and outcomes, and relate type I to hereditary nonpolyposis colorectal carcinoma (Lynch syndrome).
- Recognize the features and molecular alterations of uterine carcinosarcoma (malignant mixed Müllerian tumor).
- Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord–stromal cells) as the basis for tumor classification.
- Identify risk factors, hereditary cancer syndromes, and molecular mechanisms underlying epithelial, sex cord–stromal, and germ cell ovarian tumors.

Question 1:

A 30-year-old woman undergoing evaluation for infertility has an endometrial biopsy performed. Histologic examination reveals a thickened endometrium with sacculated glands containing secretory material and coiled spiral arteries. The glands show a saw-tooth appearance and are distended with glycogen-rich secretions.

Which of the following best describes the phase of the ovarian cycle and the corresponding days of the menstrual cycle?

- A. Follicular phase; days 6–13
- B. Ovulatory phase; days 13–15
- C. Luteal phase; days 15–28
- D. Menstrual phase; days 1–5
- E. Proliferative phase; days 6–14

Ovary: Structure and Function

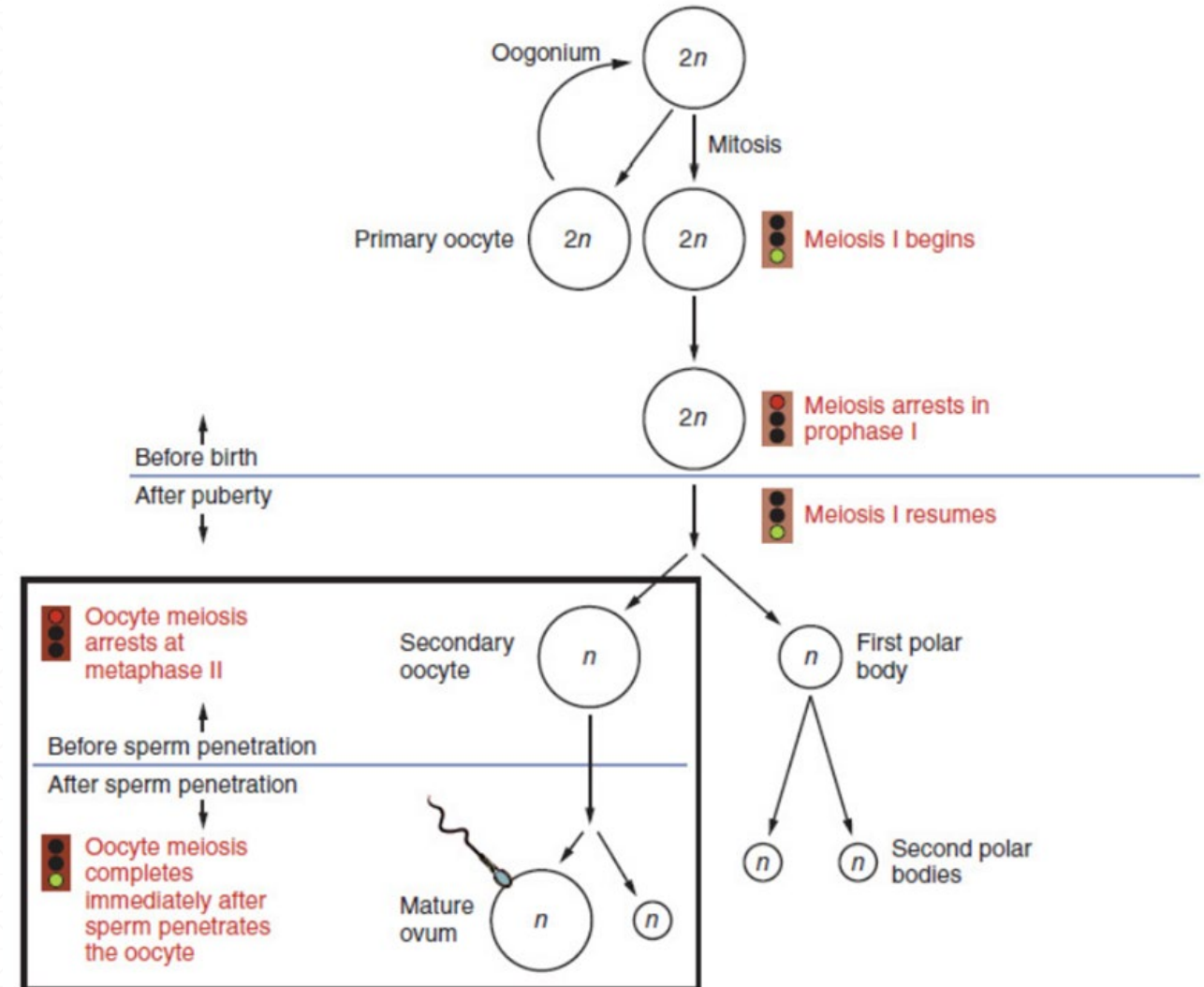
LO: Describe the microscopic organization of the ovary, uterine tube, and uterus
Relate histological changes to the ovarian cycle, menstrual cycle, and pregnancy

Microscopic organization:

- **Cortex:** Contains follicles at various stages of development, the corpus luteum, and corpora albicantia, embedded in a cellular stroma.
- **Medulla:** Loose connective tissue carrying blood vessels, lymphatics, and nerves.
- **Tunica albuginea:** Dense connective tissue layer beneath the surface epithelium.
- **Surface epithelium:** Simple cuboidal to squamous mesothelium (historically misnamed "germinal epithelium").

Oogenesis:

- Begins in fetal life (~11–12 weeks' gestation); oogonia enter meiosis I and arrest as **primary oocytes in prophase I** until puberty.
- The **LH surge resumes meiosis I**, yielding a secondary oocyte plus first polar body; the secondary oocyte is released at ovulation and **arrests in metaphase II**.
- **Meiosis II completes only if fertilization occurs**, extruding the second polar body.



Folliculogenesis:

LO: Describe the microscopic organization of the ovary, uterine tube, and uterus, and relate histological changes to the ovarian cycle, menstrual cycle, and pregnancy

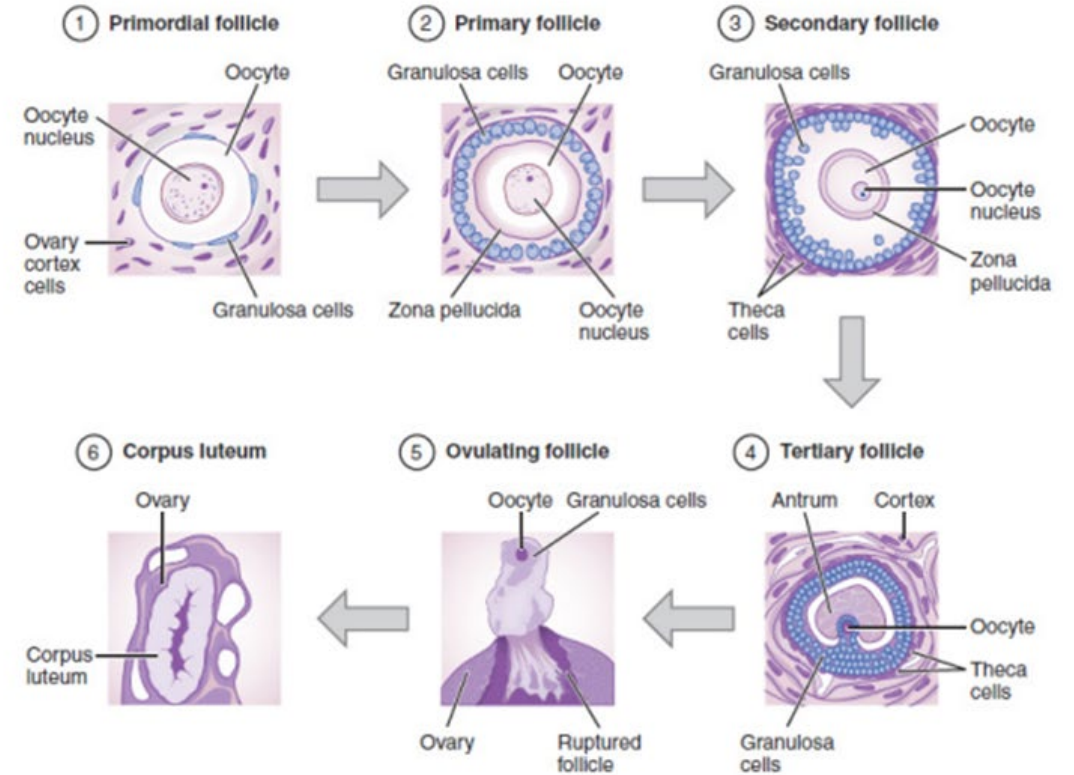
- **Primordial follicles:** Primary oocyte surrounded by a single layer of flat (squamous) pregranulosa cells. **These represent the largest pool of oocytes, formed in fetal life and held dormant until recruited or lost to atresia.**

Primary follicles: Cuboidal granulosa cells; zona pellucida forms.

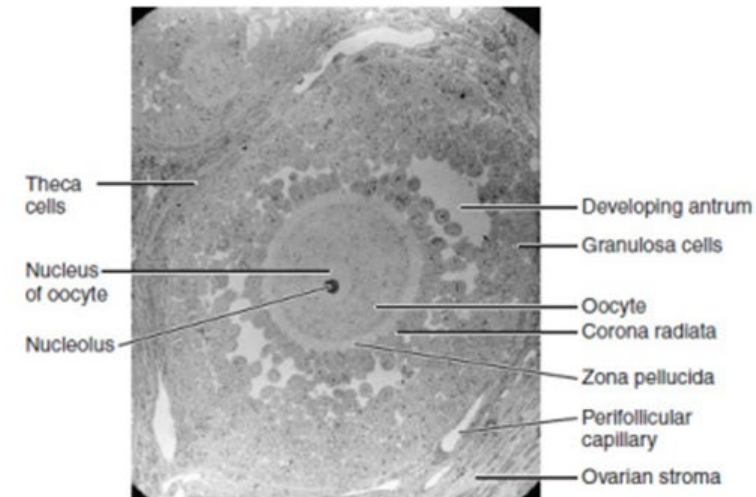
Secondary (preantral) follicles: Multiple granulosa layers; antrum begins to form; theca interna and externa differentiate.

Tertiary (antral/Graafian) follicles: Large antrum; oocyte surrounded by corona radiata on the cumulus oophorus; ready for ovulation.

(a) Stages of Folliculogenesis

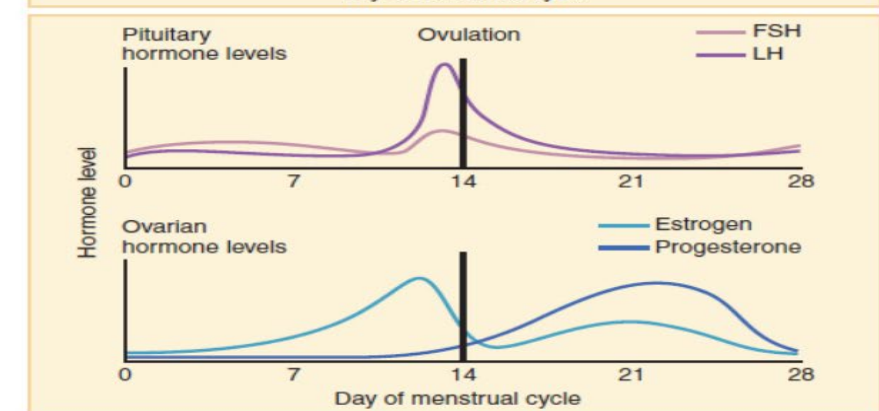
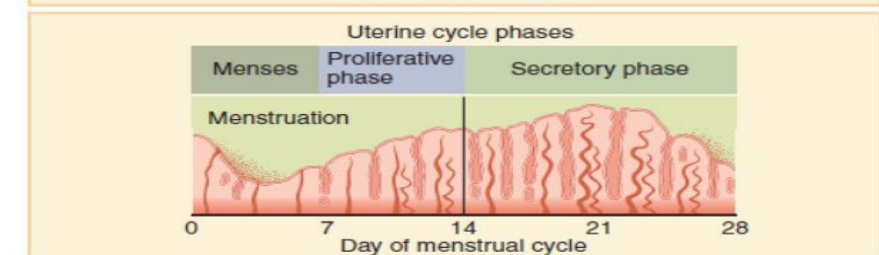
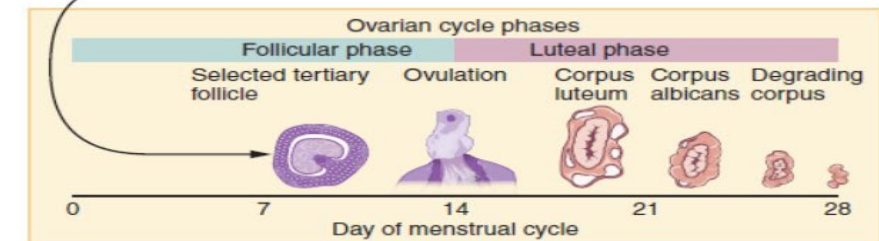
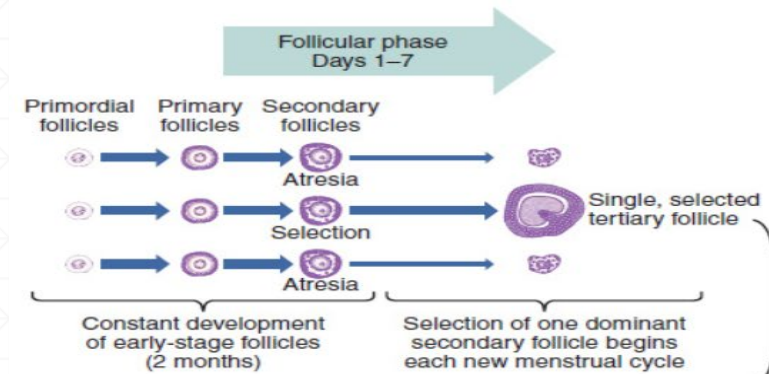


(b) A Secondary Follicle



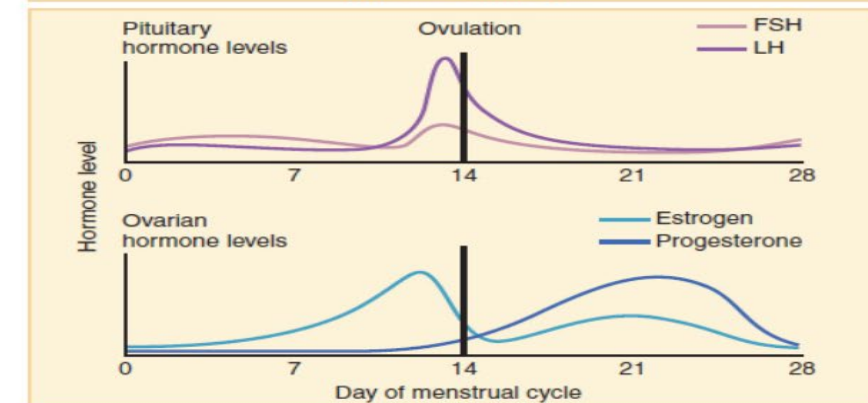
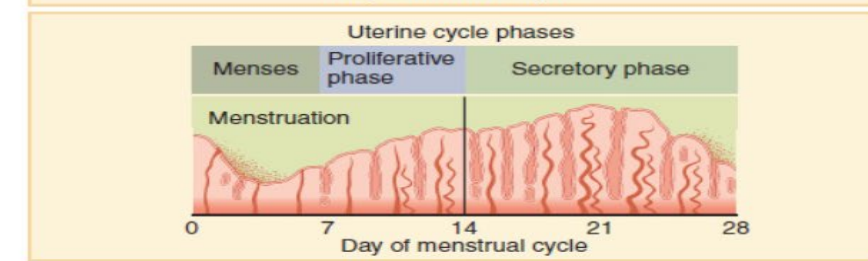
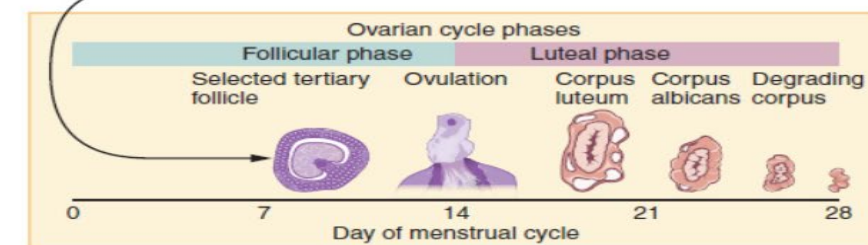
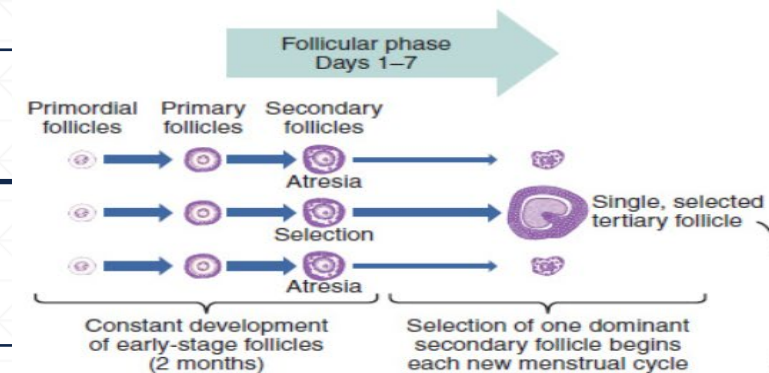
LO: Describe the microscopic organization of the ovary, uterine tube, and uterus, and relate histological changes to the ovarian cycle, menstrual cycle, and pregnancy

Cycle Phase	Days	Key Events	Histological Features	Hormonal Control	Ovarian Phase
Menses	1–5	Progesterone withdrawal → spiral artery constriction → ischemia → shedding of stratum functionalis	Sloughing tissue, extravasated blood	↓ Progesterone due to corpus luteum degeneration; ↑ FSH & LH	Late Luteal → Follicular Start corpus luteum → corpus albicans
Proliferative Phase	6–14	Estrogen-driven regeneration of stratum functionalis	Straight glands, mitotic activity, thin endometrium	FSH stimulates follicle growth → Estrogen production	Follicular Phase (variation of menstrual cycle)
Ovulation	~Day 14	LH surge triggered by high estrogen → release of secondary oocyte	Transition between proliferative and secretory phases	High estrogen → LH surge	Ovulation
Secretory Phase	14–28	Progesterone from corpus luteum → glandular secretion, spiral artery coiling	Sacculated glands, saw-tooth appearance, predecidual cells	Corpus luteum secretes progesterone; if no fertilization → ↓ progesterone → menstruation	Luteal Phase



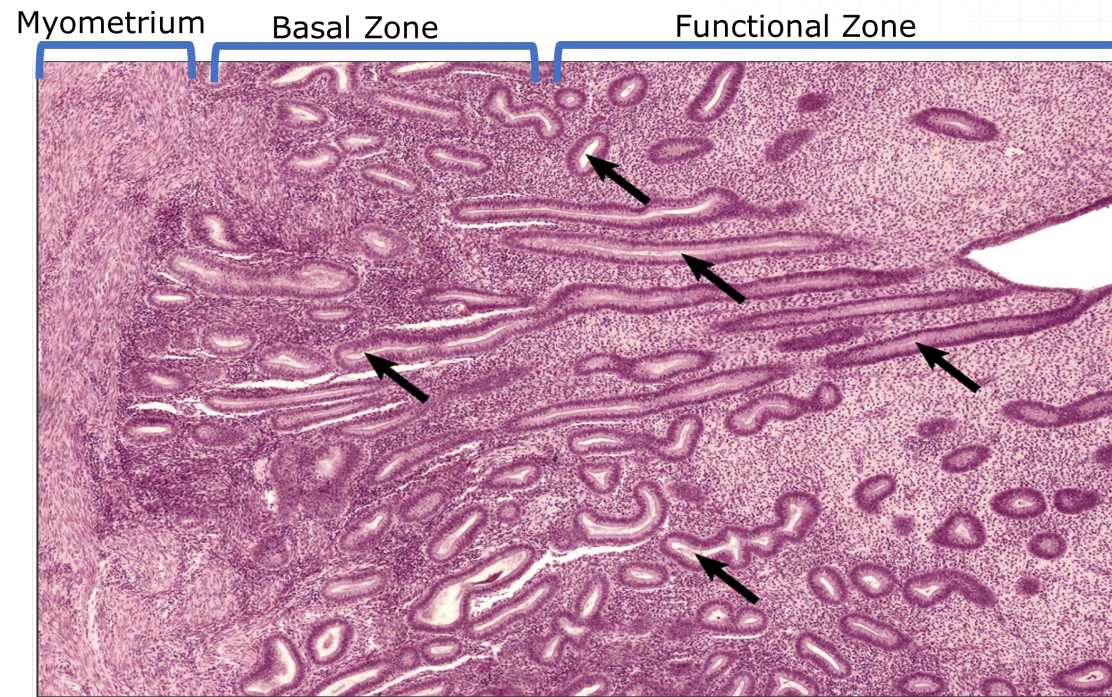
LO: Describe the microscopic organization of the ovary, uterine tube, and uterus, and relate histological changes to the ovarian cycle, menstrual cycle, and pregnancy-Continued

Ovarian cycle stage	Ovarian histology	Matching menstrual (endometrial) phase	Endometrial histology	Dominant hormonal control
Early follicular phase (primordial → primary follicles recruited)	Primordial follicles (flat granulosa) activate to primary follicles (cuboidal granulosa, zona pellucida forming); coincides with menses	Menstrual phase (days ~1–5), then early proliferative	Functional layer shed; ischemia/apoptosis with spiral-artery vasospasm, then basal-layer regeneration	Rising FSH (loss of negative feedback as prior corpus luteum regresses); low estrogen/progesterone
Mid-follicular phase (secondary/antral follicle growth, dominant follicle selection)	Antrum enlarges; theca interna secretes androstenedione; granulosa aromatize to estradiol; one dominant follicle emerges	Proliferative phase	Glandular and stromal proliferation, mitoses; straight tubular glands; spiral arteries lengthen	FSH stimulates aromatase; rising estradiol (negative feedback on FSH)
Late follicular phase (preovulatory Graafian follicle)	Large Graafian follicle; oocyte with corona radiata on cumulus oophorus	Late proliferative phase	Maximal glandular proliferation; subnuclear glycogen vacuoles appear; pseudostratified epithelium	Peak estradiol → positive feedback triggering LH surge
Ovulation (~day 14)	LH surge → follicle rupture; oocyte completes meiosis I to secondary oocyte (arrests in metaphase II) and is released	Transition to secretory phase	Onset of secretory transformation	LH surge (with FSH surge)
Early–mid luteal phase (corpus luteum)	Ruptured follicle luteinizes into corpus luteum; luteinized granulosa/theca cells	Early–mid secretory phase	Tortuous, sacculated glands; subnuclear then luminal secretion; stromal edema; coiled spiral arteries; decidualization near implantation window	LH-driven corpus luteum secretes progesterone + second estradiol rise; progesterone dominant
Late luteal phase (corpus luteum regression, if no pregnancy)	Corpus luteum → corpus albicans (scar)	Late secretory / premenstrual	Glandular breakdown, stromal apoptosis, decreased endometrial height, spiral-artery constriction	Falling estrogen and progesterone; rising FSH begins next cycle
Luteal rescue (if fertilization/implantation)	Corpus luteum maintained (corpus luteum of pregnancy)	Decidualized secretory endometrium sustained	Full decidualization; no shedding	hCG from trophoblast sustains corpus luteum → continued progesterone

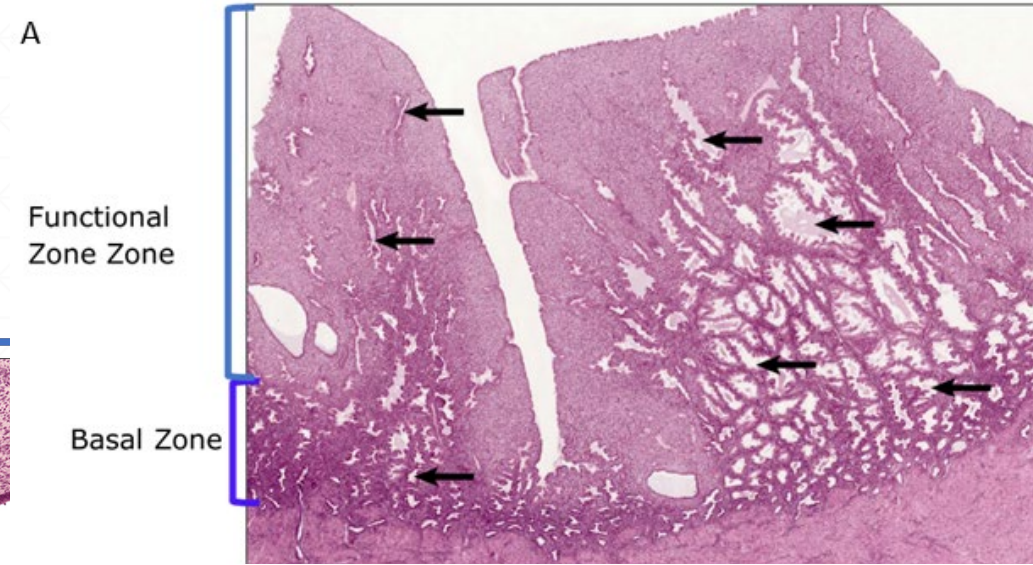


Uterine Histological Changes in

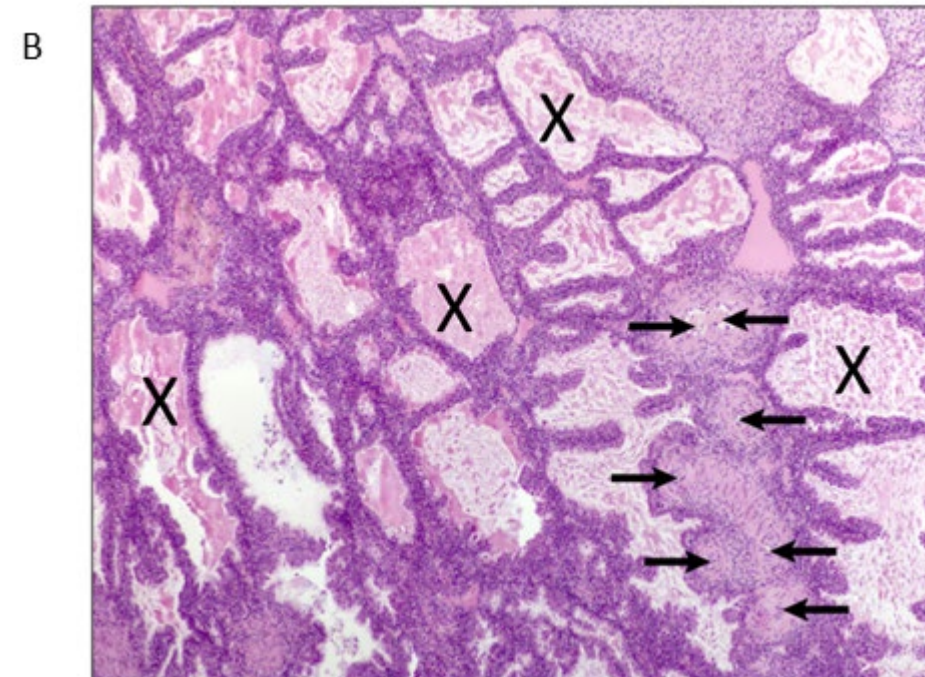
LO: Describe the microscopic organization of the ovary, uterine tube, and uterus, and relate histological changes to the ovarian cycle, menstrual cycle, and pregnancy



Proliferative endometrium. Straight, non-secretory endometrial glands



Early secretory phase endometrium with secretory products starting to accumulate in some glands (arrows)



Late secretory phase endometrium; glands are widely sacculated, with characteristic saw-tooth appearance, and lumens distended with secretions (X).

Question 2

A 30-year-old woman has had recurring pelvic pain with dysuria and dysmenorrhea for the past 5 years. On pelvic examination there are no abnormal findings. Magnetic resonance imaging shows focal, less than 3 mm, bright lesions in the anterior cul-de-sac along with adhesions. She receives therapy with a combination of oral contraceptive agent along with an aromatase inhibitor, and her condition improves. Her lesions are most likely to have which of the following microscopic appearances?

- A. Endometrial glands with stroma
- B. Enlarged, avascular chorionic villi
- C. Neutrophilic abscesses
- D. Noncaseating granulomata
- E. Poorly differentiated adenocarcinoma

Endometriosis

LO: Explain how altered structure/function leads to clinical consequences (e.g., follicular cysts, tubal infertility, endometriosis, abnormal uterine bleeding, menopause).

Endometriosis is defined by the presence of endometrial glands and stroma in a location outside the uterus

- Multifocal and involves pelvic structures (e.g., ovaries, pouch of Douglas, uterine ligaments, fallopian tubes)
- Less frequently, involves distant areas of the peritoneal cavity, periumbilical tissues, or laparotomy scars
- **There are three types of endometriosis:**
 - superficial peritoneal endometriosis
 - ovarian endometriosis
 - deep infiltrating endometriosis
- **Risk of malignant transformation is mainly confined to deep infiltrating endometriosis**
 - lesions that penetrate more than 5 mm beneath the peritoneal surface, often involving structures like the bowel, bladder, and uterosacral ligaments.
 - More prone to genetic mutations and cellular changes that may lead to malignancy

Endometriosis-Pathogenesis

LO: Explain how altered structure/function leads to clinical consequences (e.g., follicular cysts, tubal infertility, endometriosis, abnormal uterine bleeding, menopause).

Pathogenesis

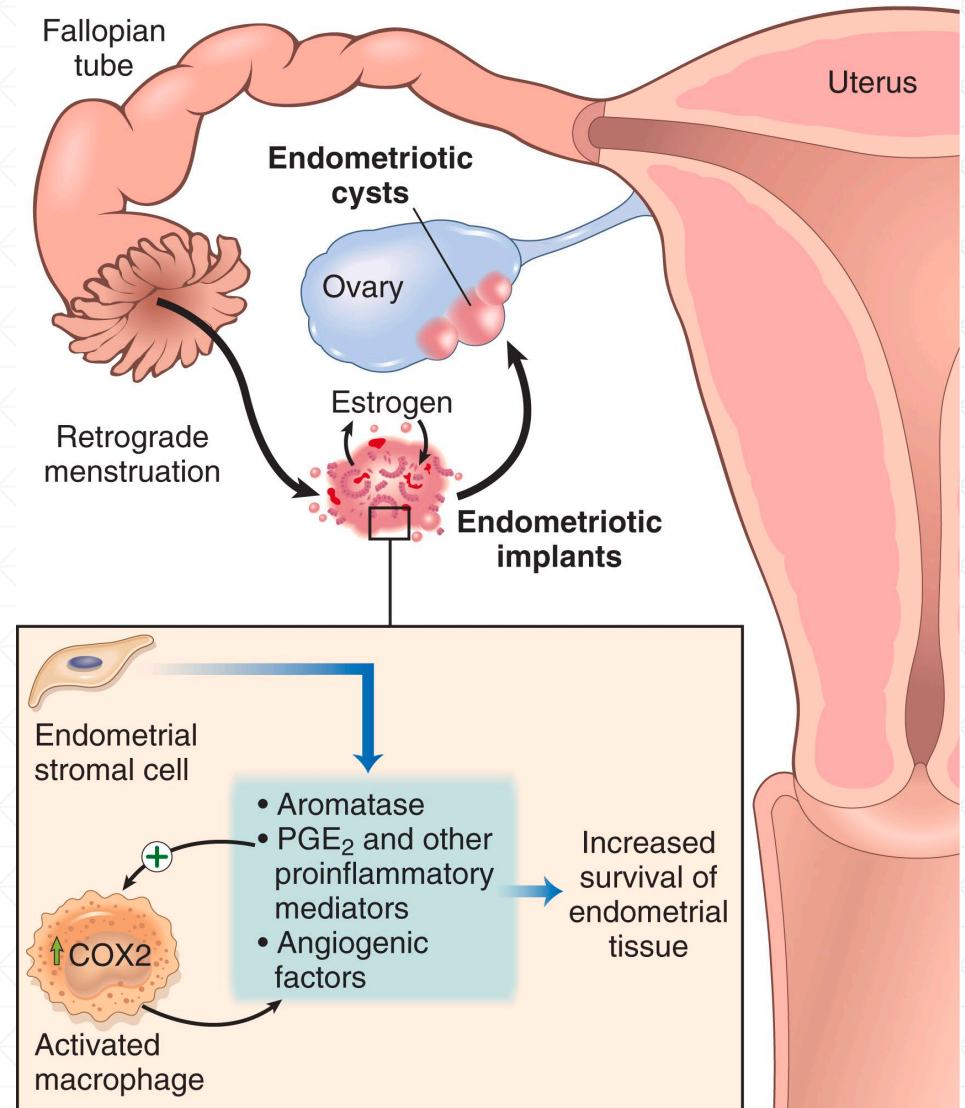
1. origin from the uterine endometrium
2. origin from cells outside the uterus giving rise to endometrial tissue

The leading theories are as follows:

- The regurgitation theory: endometrial tissue implants at ectopic sites via retrograde flow of menstrual endometrium through the opening of the fallopian tube
- The benign metastasis theory: endometrial tissue can “spread” to distant sites (e.g., bone, lung, and brain) via blood vessels and lymphatic channels
- The metaplastic theory: endometrium arises directly from coelomic epithelium (mesothelium of pelvis or abdomen), from which the müllerian ducts and ultimately the endometrium originate during embryonic development. Mesonephric remnants may undergo endometrial differentiation and give rise to ectopic endometrial tissue.
- The extrauterine stem/progenitor cell theory: stem/progenitor cells from the bone marrow differentiate into endometrial tissue.

Implants have different characteristics than eutopic uterine endometrium

- Endometriotic tissue exhibits increased levels of proinflammatory and angiogenic factors, including prostaglandin E_2 , vascular endothelial growth factor (VEGF), and matrix metalloproteinases (MMP), some of which are released by macrophages that are recruited to endometriotic implants by proinflammatory factors
- Endometriotic stromal cells make high levels of aromatase, leading to increased local production of estrogen from androgens



Female Genital System and Breast

Kumar, Vinay, MBBS, MD, FRCPath, Robbins & Kumar Basic Pathology, 17, 602-635

Pathogenesis of endometriosis. Depicted is the interplay between factors expressed in endometriotic implants and activated macrophages that are hypothesized to play a role in the establishment and maintenance of endometriotic implants. COX2, Cyclo...

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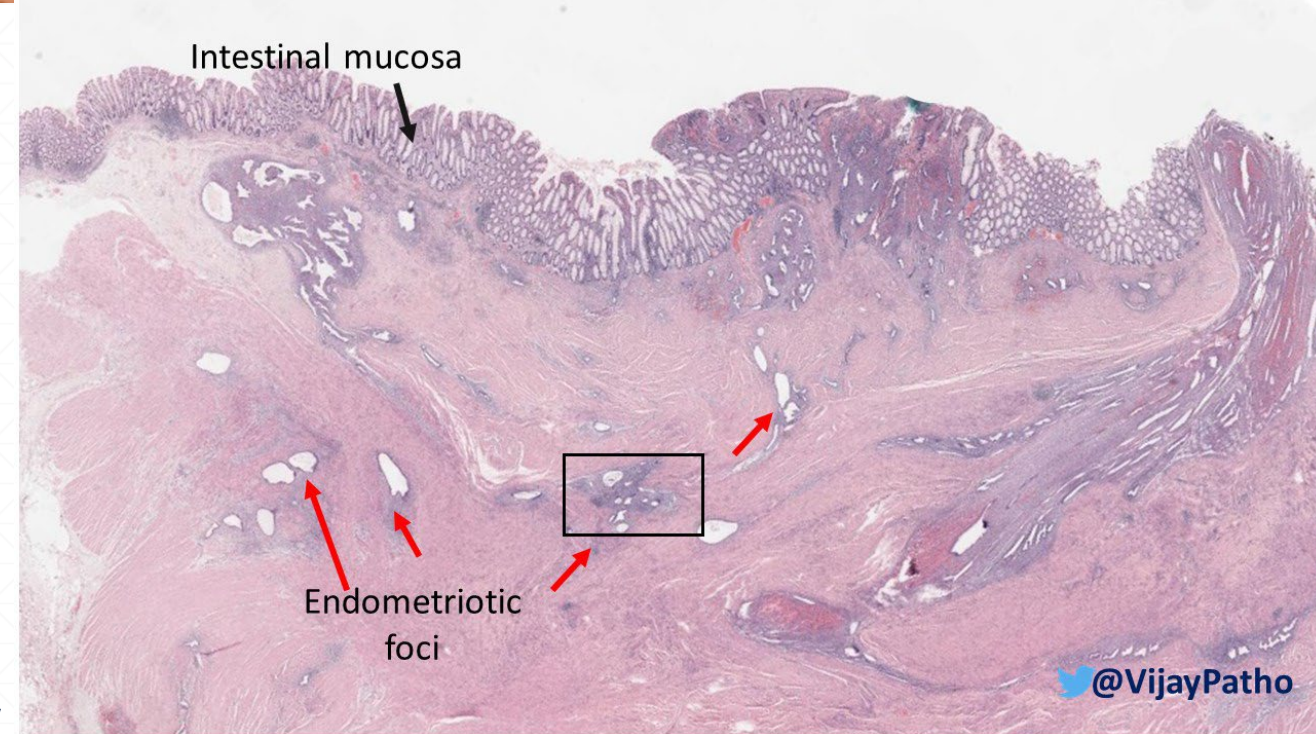
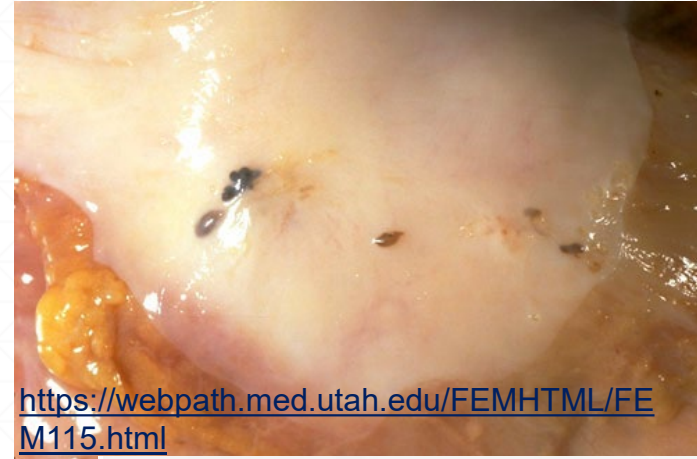
Endometriosis-Morphology

LO: Explain how altered structure/function leads to clinical consequences (e.g., follicular cysts, tubal infertility, endometriosis, abnormal uterine bleeding, menopause)

Explain the pathophysiology and clinicopathologic features of functional disorders, including polycystic ovary syndrome, endometrial hyperplasia, adenomyosis, and endometriosis.

Morphology

- Consists of **functioning endometrium**; undergoes cyclic bleeding
- Collection of blood in aberrant foci---**appear red-brown nodules or implants, ranging in size from microscopic to 1 to 2 cm in diameter, that lie on or just under the affected serosal surface**
- When lesions are widespread, organizing hemorrhage can cause extensive fibrous adhesions between fallopian tubes, ovaries, and other structures and obliterate the pouch of Douglas
- The ovaries may become markedly distorted by large cystic masses (3 to 5 cm in diameter) filled with brown fluid resulting from previous hemorrhage; **chocolate cysts** or **endometriomas**
- **Diagnosis depends on finding both endometrial glands and stroma at sites external to the endometrium**



Question 3:

A 42-year-old woman presents with heavy menstrual bleeding and pelvic pain that worsens just before menstruation. Physical examination reveals a mildly enlarged, tender uterus. Transvaginal ultrasound shows a thickened uterine wall with a globular contour. Histologic examination of the uterine wall reveals endometrial glands and stroma embedded within the myometrium.

Which of the following best explains the pathophysiology underlying this patient's presentation?

- A. Ectopic endometrial tissue induces myometrial hypertrophy
- B. Unopposed estrogen stimulation of the endometrium
- C. Ovulatory dysfunction leading to anovulation
- D. Endometrial hyperplasia due to chronic inflammation
- E. Submucosal fibroid disrupting endometrial architecture

Adenomyosis

LO: Explain the pathophysiology and clinicopathologic features of functional disorders, including polycystic ovary syndrome, endometrial hyperplasia, adenomyosis, and endometriosis.

Definition

Adenomyosis is the presence of endometrial glands and stroma within the myometrium (uterine muscle wall)

Pathophysiology

- Endometrial tissue becomes embedded deep within the myometrium between muscle bundles
- This ectopic tissue induces reactive hypertrophy of surrounding smooth muscle
- Results in a globular, enlarged uterus with a thickened uterine wall

Clinical Features

- Menorrhagia (heavy menstrual bleeding)
- Dysmenorrhea (painful menstruation)
- Pelvic pain, especially pre-menstrual
- May coexist with endometriosis, but they are distinct entities.

Gross and Imaging Findings

- Uterus appears enlarged and boggy
- Wall may be diffusely thickened
- Diagnosed via MRI or ultrasound, though definitive diagnosis may require histology



<https://www.pathologyoutlines.com/topic/uterusadenomyosis.html>



<https://ilovepathology.com/pathology-of-adenomyosis/>

Question 4:

A study of patients with postmenopausal uterine bleeding reveals that some have malignant neoplasms arising from prior atypical hyperplastic lesions. These neoplasms are most common in women aged 55–65 with obesity, hypertension, or diabetes mellitus. Histologic examination shows glandular architecture resembling proliferative endometrium. Molecular analysis reveals frequent PTEN mutations and occasional mismatch repair gene mutations. The tumors are typically well-differentiated and remain localized for years before spreading via lymphatics.

Which of the following neoplasms is most consistent with these findings?

- A. Clear cell carcinoma
- B. Endometrioid carcinoma
- C. Leiomyosarcoma
- D. Müllerian mixed tumor (carcinosarcoma)
- E. Serous carcinoma
- F. Stromal sarcoma

Endometrial Hyperplasia

LO: Explain the pathophysiology and clinicopathologic features of functional disorders, including polycystic ovary syndrome, endometrial hyperplasia, adenomyosis, and endometriosis

Definition & Pathogenesis

Endometrial hyperplasia is an abnormal proliferation of endometrial glands due to **excess estrogen unopposed by progesterone**

- It is a precursor lesion for Type I endometrial carcinoma (endometrioid type).

Etiologies of Estrogen Excess

- Obesity (peripheral conversion of androgens to estrogens in adipose tissue)
- Polycystic ovary syndrome (PCOS)
- Estrogen-producing ovarian tumors (e.g., granulosa cell tumors)
- Exogenous estrogen therapy without progestin

Classification

1. Hyperplasia without atypia

- Increased gland-to-stroma ratio
- Glands may be cystically dilated
- Low risk of progression to carcinoma (1–3%)

2. Hyperplasia with atypia (also called Endometrial Intraepithelial Neoplasia, EIN)

- Complex glandular architecture
- Nuclear atypia: enlarged, vesicular nuclei, prominent nucleoli
- High risk of progression to carcinoma
- Associated with PTEN mutations

Endometrial Hyperplasia-Morphology

LO: Explain the pathophysiology and clinicopathologic features of functional disorders, including polycystic ovary syndrome, endometrial hyperplasia, adenomyosis, and endometriosis

Histological Features

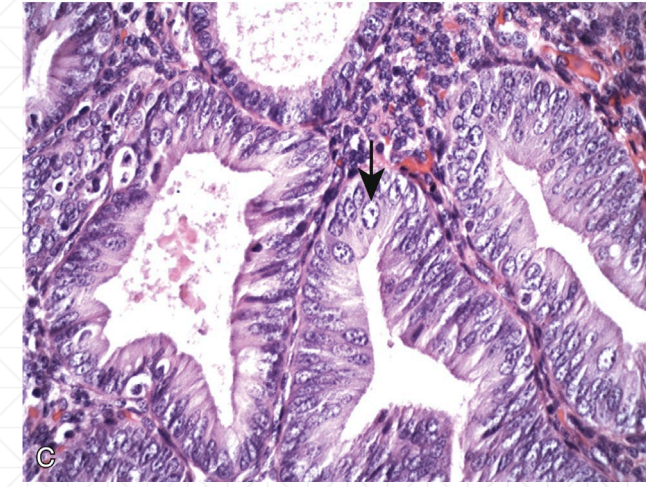
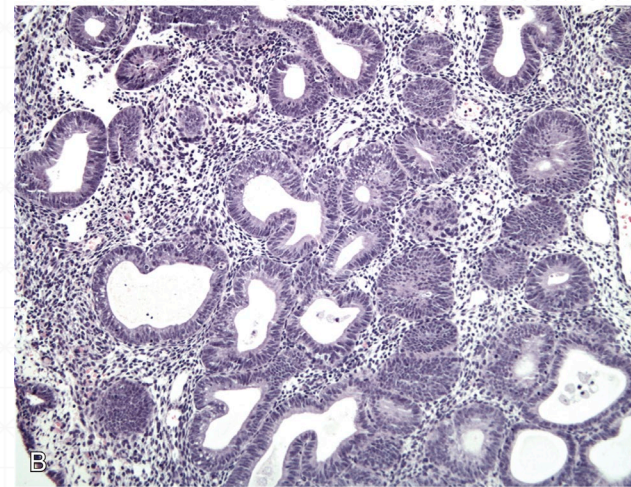
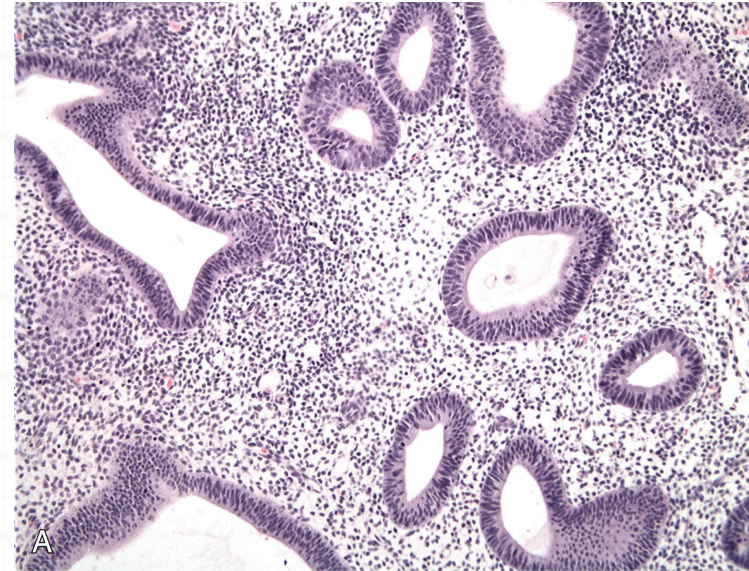
- **Without atypia; A:** mild glandular crowding, preserved stroma
- **With atypia; B and C:** crowded glands, reduced stroma, cytologic atypia
- PTEN mutation is a shared molecular feature with endometrioid carcinoma

Clinical Implications

- Presents with abnormal uterine bleeding, especially in perimenopausal or postmenopausal women
- Requires careful evaluation to exclude invasive carcinoma

Management:

- Hysterectomy for patients not desiring fertility
- High-dose progestins for fertility preservation in younger patients



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Endometrial Carcinoma-Endometrioid

LO: Differentiate between type I and type II endometrial carcinomas in terms of risk factors, molecular basis, morphology, and outcomes, and relate type I to hereditary nonpolyposis colorectal carcinoma (Lynch syndrome).

Type I Endometrial Carcinoma (Endometrioid Type)

Risk Factors:

- Mutations in mismatch repair genes and the tumor suppressor gene *PTEN* are early events in the stepwise development of endometrioid carcinoma.
- Estrogen excess: obesity, PCOS, estrogen-producing tumors, unopposed estrogen therapy
- Associated conditions: endometrial hyperplasia with atypia (EIN)
- Age group: perimenopausal women (peak 55–65 years)

Molecular Basis:

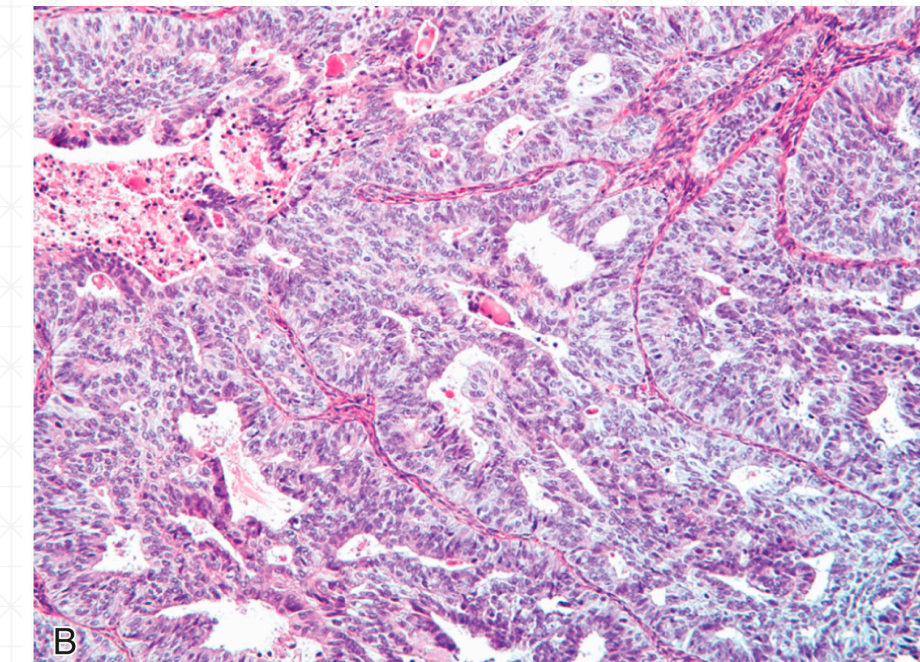
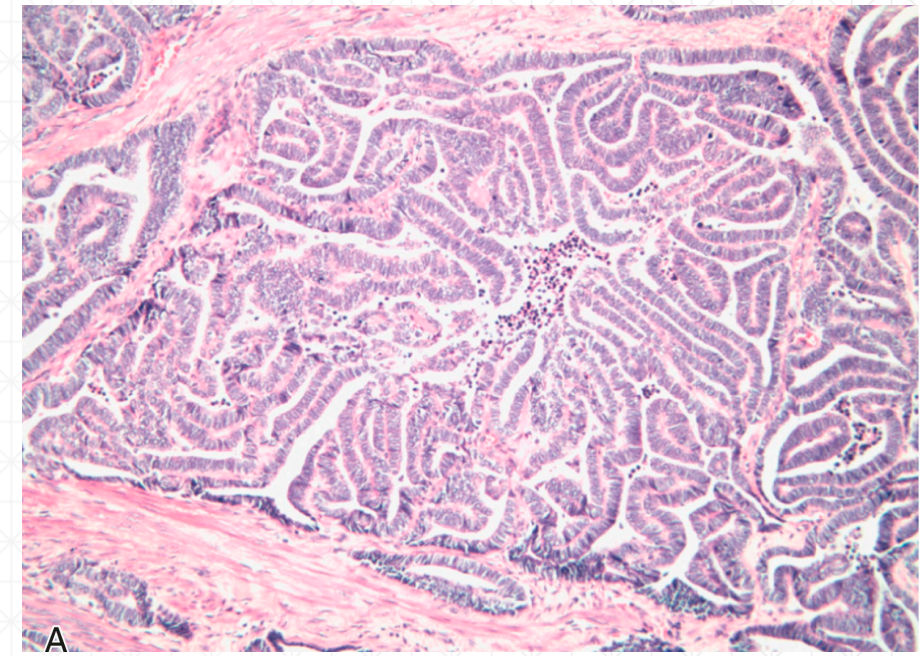
- *PTEN* mutations (Cowden syndrome)
- Mismatch repair gene mutations (e.g., Lynch syndrome)
- Occasional *TP53* mutations (late event)

Morphology:

- Glandular architecture resembling proliferative endometrium
- Graded 1–3 based on differentiation:
 - Grade 1; A: well-differentiated glands
 - Grade 3; B: solid growth, marked atypia

Clinical Features:

- Presents with postmenopausal bleeding
- Slow progression, often localized
- Good prognosis if detected early (90% 5-year survival in early stages)



Endometrial Carcinoma-Serous

LO: Differentiate between type I and type II endometrial carcinomas in terms of risk factors, molecular basis, morphology, and outcomes, and relate type I to hereditary nonpolyposis colorectal carcinoma (Lynch syndrome).

Type II Endometrial Carcinoma (Serous Type):

Risk Factors:

- Endometrial atrophy, not estrogen-related
- Older age: typically 65–70 years
- Not associated with obesity or hyperplasia

Molecular Basis:

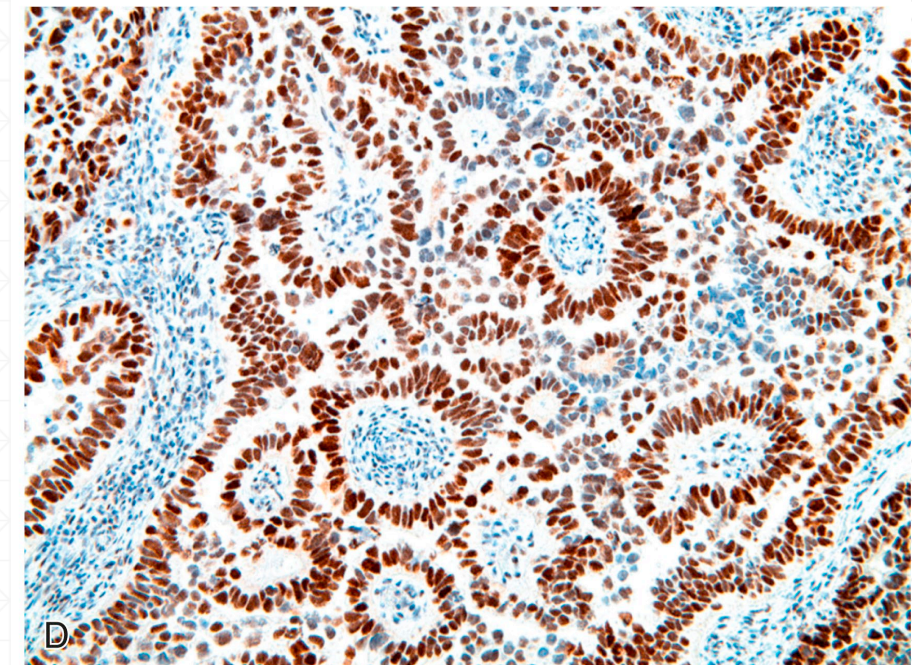
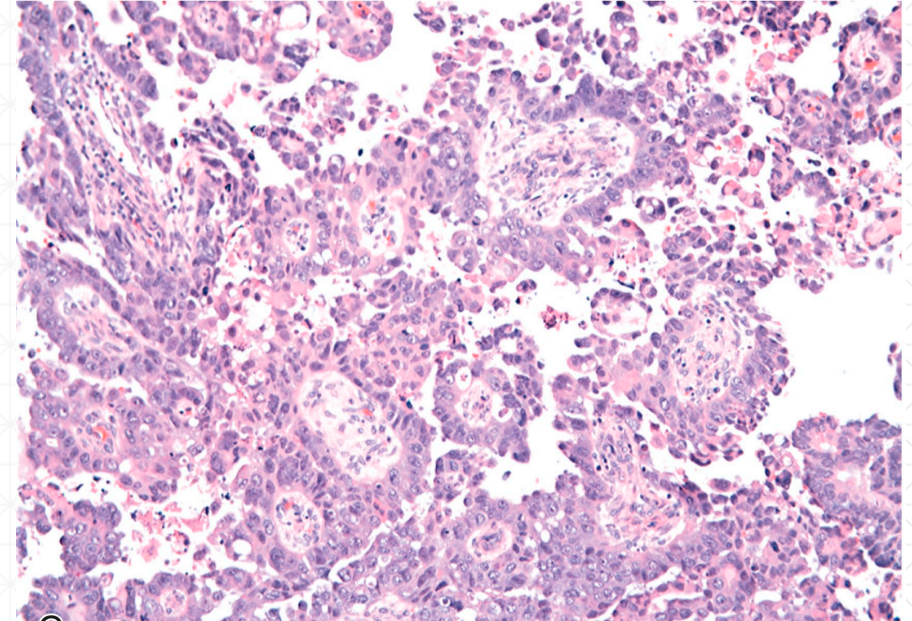
- TP53 mutations (early and central event)
- Serous tumors are preceded by serous endometrial intraepithelial carcinoma (SEIC), in which *TP53* mutations are often detected, indicating a central role for altered p53 function in the development of this form of endometrial carcinoma, which shows significant morphologic and biologic overlap with ovarian serous carcinoma.

Morphology:

- Papillary structures, marked cytologic atypia; C: High-grade by definition
- Diffuse p53 immunostaining due to mutant protein accumulation; D

Clinical Features:

- Aggressive behavior
- Often presents at advanced stage
- Poor prognosis, dependent on staging



Question 5:

A 52-year-old female is having her annual physical examination. Her only complaint is occasional heavy menstrual periods. Her gynecologist palpated a mass in the uterus. She underwent a hysterectomy. Examination of the resected uterus revealed the pathologic lesion illustrated in the figure. Of the following, which is most characteristic of this condition?

- A. Malignant transformation is common.
- B. It is a very rare cause of infertility.
- C. Microscopic examination would reveal interlocking fascicles of skeletal muscle.
- D. The condition is estrogen sensitive.
- E. The condition is almost exclusively present in white females.



Question 6:

A 52-year-old woman presents with abnormal uterine bleeding and pelvic pain that has progressively worsened over the past six months. On physical examination, an irregularly shaped mass is palpated in the lower abdomen. Pelvic ultrasound reveals a large, heterogeneous uterine mass. A biopsy is performed, and histologic examination shows spindle-shaped cells with marked nuclear atypia, high mitotic activity, and areas of necrosis. Which of the following is the most likely diagnosis?

- A. Endometrial hyperplasia
- B. Adenomyosis
- C. Leiomyoma
- D. Leiomyosarcoma
- E. Endometrial carcinoma

Uterine Leiomyoma (Fibroid)

LO: Compare benign and malignant uterine tumors (e.g., leiomyomas vs. leiomyosarcomas), including morphology, clinical course, and prognosis.

- Most common tumors in women
- Benign smooth muscle neoplasms; single or multiple
- Estrogen and progesterone stimulate leiomyoma growth; tumors shrink after menopause
- Associated with recurrent chromosomal abnormalities, including rearrangements of chromosomes 6 and 12 found in other benign neoplasms (e.g., endometrial polyps and lipomas)
- Mutations in the MED12 gene; encodes a protein that regulates RNA polymerase II-mediated transcription, have been identified in up to 70% of leiomyomas
- The mechanism by which MED12 mutations contribute to the development of leiomyomas is not understood.

Uterine Leiomyoma (Fibroid)-Morphology

LO: Compare benign and malignant uterine tumors (e.g., leiomyomas vs. leiomyosarcomas), including morphology, clinical course, and prognosis.

Gross:

- Sharply circumscribed, firm gray-white masses with a characteristic whorled cut surface
- Ranging from small nodules to large tumors
- Types: embedded within the myometrium (intramural), lie beneath the endometrium (submucosal) or the serosa (subserosal; may extend out on attenuated stalks (pedunculated)
- Subserosal may become attached to surrounding organs and develop a blood supply (parasitic leiomyomas)
- Submucosal leiomyomas may ulcerate and cause abnormal uterine bleeding (menometrorrhagia)
- **Submucosal leiomyomas are associated with increased risk for spontaneous abortion**
- Malignant transformation to leiomyosarcoma is extremely rare

Histologic examination:

- Bundles of intersecting uniform smooth muscle cells (elongated nuclei and eosinophilic cytoplasm) similar in appearance to typical myometrium
- Foci of fibrosis, calcification, and degenerative softening may be present

Leiomyosarcoma

LO: Compare benign and malignant uterine tumors (e.g., leiomyomas vs. leiomyosarcomas), including morphology, clinical course, and prognosis.

- Uncommon malignant neoplasms
- Arise from the myometrium or endometrial stromal precursor cells
- **Almost always solitary and most often occur in postmenopausal women**, in contrast to leiomyomas; leiomyomas are frequently multiple and usually arise before menopause
- They have complex, highly variable karyotypes including chromosomal deletions
- Like leiomyomas, they contain MED12 mutations but in a smaller subset (30%).

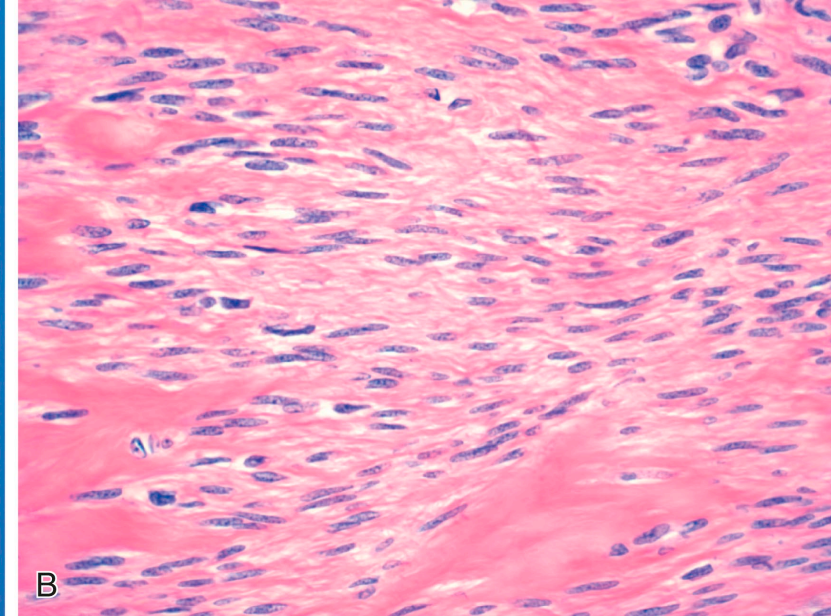
Morphology

- Soft, hemorrhagic, necrotic masses
- The histopathology is highly variable; from tumors resembling leiomyoma to wildly anaplastic neoplasms
- **The diagnostic features of leiomyosarcoma include tumor necrosis, cytologic atypia, and mitotic activity**
- Because increased mitotic activity is sometimes seen in benign smooth muscle tumors, particularly in young women, an **assessment of all three features is necessary to make a diagnosis of malignancy**

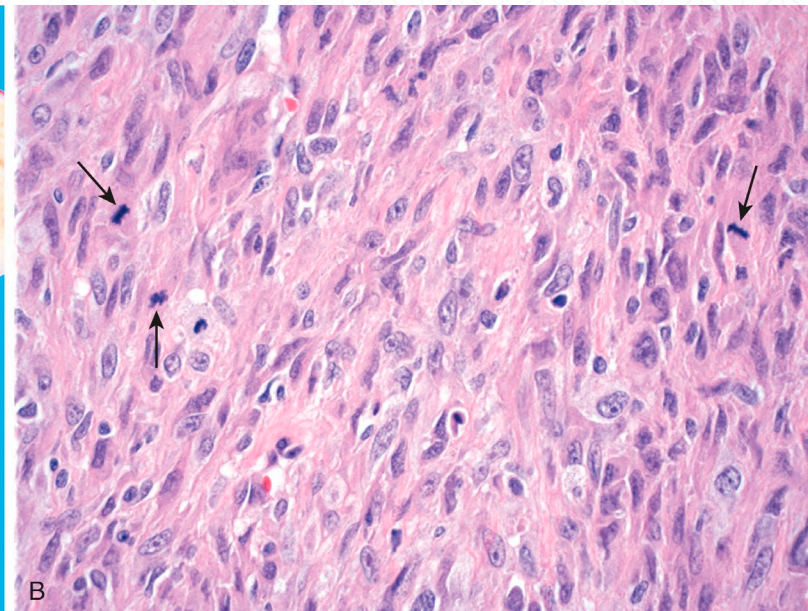
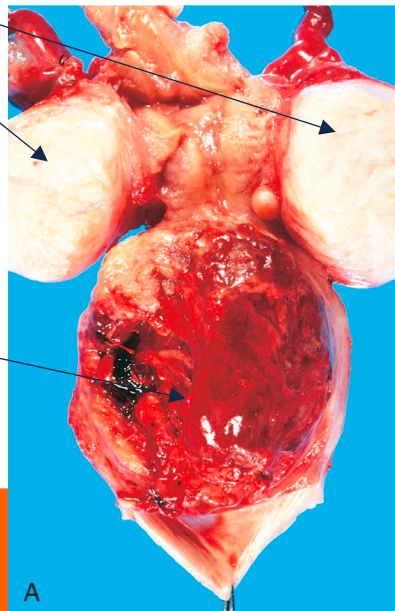
Leiomyoma vs Leiomyosarcoma

LO: Compare benign and malignant uterine tumors (e.g., leiomyomas vs. leiomyosarcomas), including morphology, clinical course, and prognosis.

Leiomyomas



Leiomyosarcoma



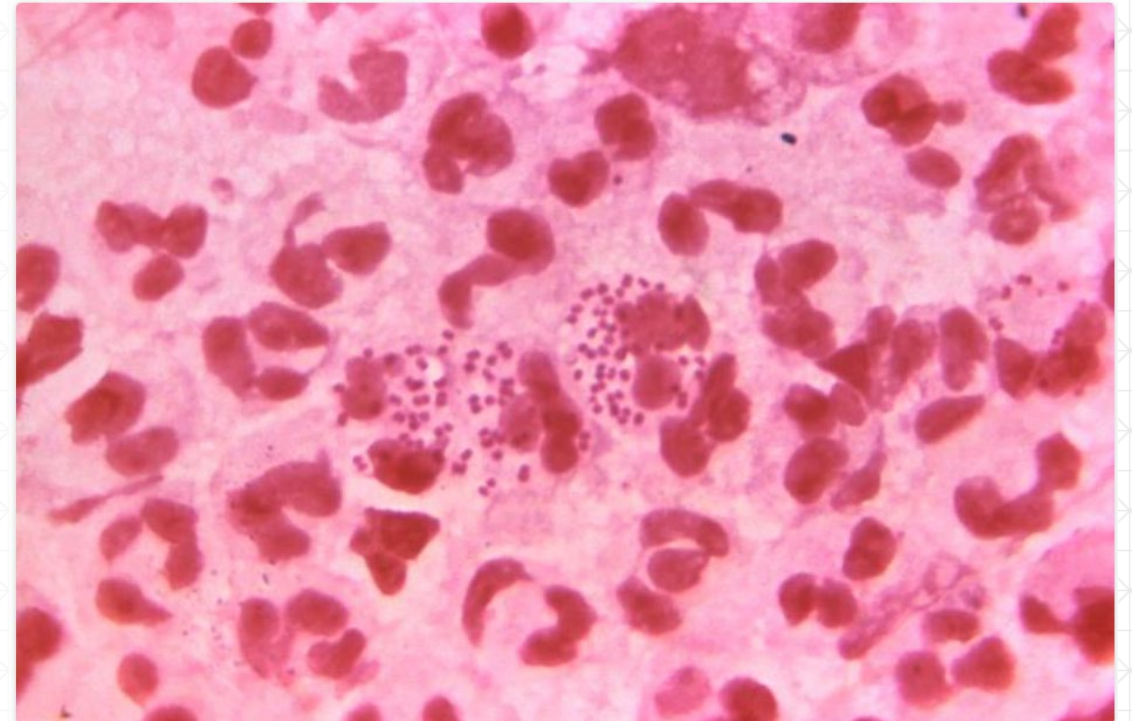
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Question 7:

A 24-year-old woman presents to the emergency department with fever, lower abdominal pain, and vaginal discharge. Pelvic examination reveals cervical motion tenderness and adnexal tenderness. Ultrasound shows bilateral adnexal masses with complex fluid collections. A laparoscopic examination reveals hemorrhagic, friable tissue and adhesions between the fallopian tubes and ovaries. A Gram stain of the cervical discharge is provided.

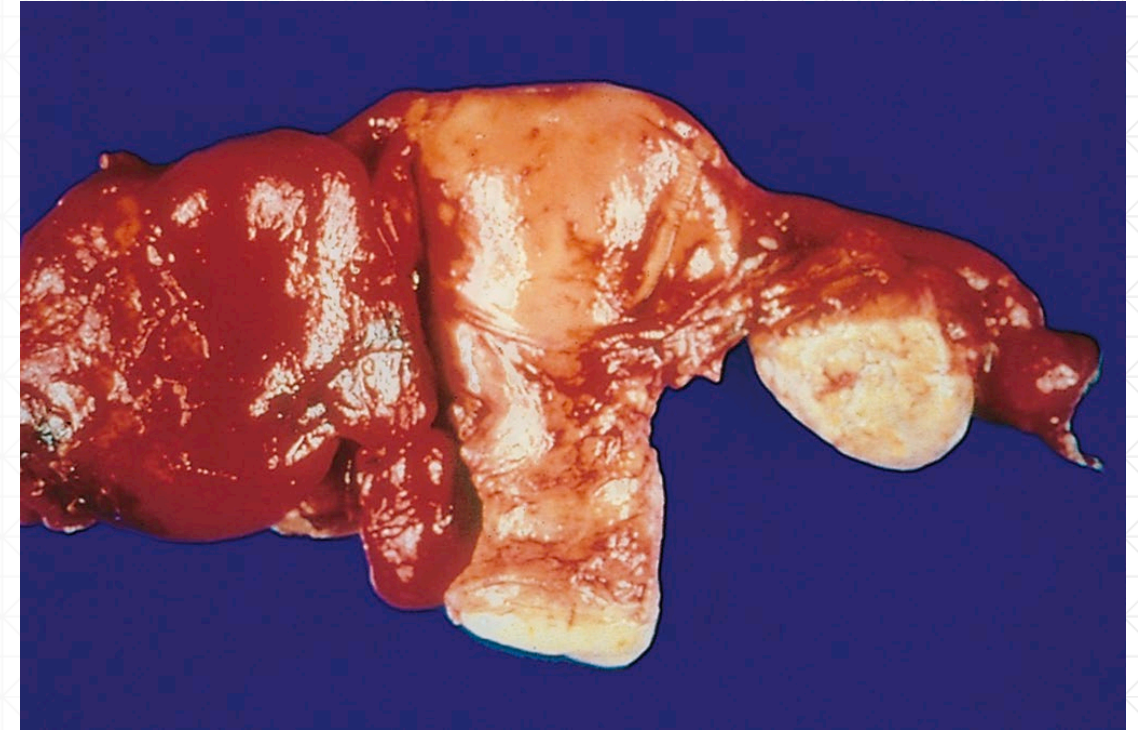
Which of the following is the most likely long-term complication of this patient's condition?

- A.** Ovarian torsion
- B.** Tubal ectopic pregnancy
- C.** Endometrial hyperplasia
- D.** Cervical carcinoma
- E.** Polycystic ovary syndrome



Inflammation of Fallopian Tubes

- Discuss pelvic inflammatory disease and infections of the uterine tubes and endometrium: common pathogens, pathogenesis, morphology, and complications.
- It is almost always caused by infection.
- Suppurative salpingitis caused by any pyogenic organism
- **N. gonorrhoeae** is the causative organism in more than 60% of cases
- **C. trachomatis** is responsible for the remaining cases.
- **Tuberculous**; rare in the USA, more common in endemic areas---- infertility.
- C/P: fever, lower abdominal or pelvic pain, and pelvic masses due to distention of the tubes with exudate, inflammatory debris, or tuboovarian abscess
- Adhesions form between the ovary and the tubes
- Adhesions between the ovary and the tubal pilcae--- increased risk of tubal ectopic pregnancy
- Damage to or obstruction of the tubal lumina results in permanent sterility



Bilateral and asymmetric PID. A hemorrhagic inflammatory mass totally obscures the tube and ovary to the left of the uterus. The tube is adherent to the adjacent ovary on the other side.

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Pelvic inflammatory disease, bilateral and asymmetric. The tube and ovary to the left of the uterus are totally obscured by a hemorrhagic inflammatory mass. The tube is adherent to the adjacent ovary on the other side. Gross specimen image showing...

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Question 8:

A 38-year-old woman with a BRCA1 mutation undergoes prophylactic salpingo-oophorectomy. Histological examination of the fimbriated end of the fallopian tube reveals atypical epithelial cells with high nuclear-to-cytoplasmic ratio and strong p53 immunostaining. No invasive carcinoma is found.

Which of the following best describes the lesion identified?

- A.** Endometrial intraepithelial neoplasia
- B.** Serous tubal intraepithelial carcinoma
- C.** Tuberculous salpingitis
- D.** Follicular cyst
- E.** Adenomyosis

Serous tubal intraepithelial carcinoma

Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord–stromal cells) as the basis for tumor classification.

Identify risk factors, hereditary cancer syndromes, and molecular mechanisms underlying epithelial, sex cord–stromal, and germ cell ovarian tumors.

1. Site of Origin for High-Grade Serous Carcinoma

- STIC lesions in the fallopian tube, particularly at the fimbriated end, are now recognized as the precursor lesions for many high-grade serous carcinomas previously thought to originate in the ovary.
- This has shifted the paradigm from “ovarian origin” to fallopian tube origin for these aggressive tumors.

2. Molecular Features

- >90% of STICs harbor TP53 mutations, similar to uterine serous carcinoma.
- These mutations are a hallmark of high-grade serous neoplasia.

3. Association with BRCA Mutations

- STICs are frequently found in prophylactically removed fallopian tubes from women with BRCA1 or BRCA2 mutations.
- Less commonly seen in women without known genetic risk factors.

4. Clinical Implications

- Prophylactic salpingo-oophorectomy in BRCA mutation carriers can detect STICs before invasive carcinoma develops.
- Early identification may prevent progression to high-grade serous carcinoma involving the ovary, omentum, and peritoneal cavity.

5. Histological Features

- Atypical epithelial cells with:
 - High nuclear-to-cytoplasmic ratio
 - Strong p53 immunostaining
 - No evidence of stromal invasion in STIC.

Question 9:

A 22-year-old woman presents with irregular menstrual cycles, acne, and increased facial hair. BMI is 32 kg/m². Pelvic ultrasound reveals bilaterally enlarged ovaries with multiple subcortical cysts. Laboratory studies show elevated LH:FSH ratio and increased serum testosterone.

Which of the following best explains the pathophysiology of this patient's condition?

- A.** Increased aromatase activity in granulosa cells
- B.** Excessive androgen production due to dysregulated steroidogenesis
- C.** Estrogen deficiency due to premature ovarian failure
- D.** Autoimmune destruction of ovarian follicles
- E.** Excessive FSH stimulation of theca cells

Polycystic Ovary Syndrome (PCOS)

LO: Explain the pathophysiology and clinicopathologic features of functional disorders, including polycystic ovary syndrome, endometrial hyperplasia, adenomyosis, and endometriosis.

Clinicopathological Features:

- A complex endocrine disorder
- Characterized by signs and symptoms of androgen excess and ovulatory dysfunction (e.g., menstrual abnormalities, hirsutism, polycystic ovaries, chronic anovulation, and decreased fertility)
- The most common endocrine/metabolic disorders of women
- First present at puberty
- The etiology of PCOS involves both environmental and genetic factors
- Dysregulation of enzymes involved in androgen biosynthesis, resulting in excessive androgen production
- **Patients are at increased risk for metabolic syndrome, type 2 diabetes, hypertension, cerebrovascular accidents, and endometrial hyperplasia and carcinoma**
- The ovaries are twice the normal size, and studded with subcortical cysts (double the no of follicles in normal ovaries); size is 0.5 to 1.5 cm in diameter
- Ovaries contain numerous 'cysts'; these are not cysts but egg-containing follicles that did not develop normally due to hormonal imbalances.
- **Thickened, fibrotic ovarian capsule overlying cystic follicles lined by granulosa cells with a hyperplastic luteinized theca interna**
- Absence of corpora lutea due to anovulation

Polycystic Ovary Syndrome (PCOS)-Clinicopathological-Cont

LO: Explain the pathophysiology and clinicopathologic features of functional disorders, including polycystic ovary syndrome, endometrial hyperplasia, adenomyosis, and endometriosis.

Clinicopathological Features:

- The ovaries are twice the normal size, and studded with subcortical cysts (double the no of follicles in normal ovaries); size is 0.5 to 1.5 cm in diameter
- Ovaries contain numerous 'cysts'; these are not cysts but egg-containing follicles that did not develop normally due to hormonal imbalances.
- **Thickened, fibrotic ovarian capsule overlying cystic follicles lined by granulosa cells with a hyperplastic luteinized theca interna**
- Absence of corpora lutea due to anovulation



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Polycystic ovarian syndrome. Hysterectomy specimen showing enlarged uterus and bilateral enlarged ovaries with multiple subcortical ovarian cysts. Courtesy of Dr. Shashidhar V. Murthy, James Cook University, Australia. Gross specimen shows enlarge...

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Polycystic Ovary Syndrome (PCOS)-Pathophysiology

LO: Explain the pathophysiology and clinicopathologic features of functional disorders, including polycystic ovary syndrome, endometrial hyperplasia, adenomyosis, and endometriosis.

Pathophysiology: (↑LH, ↓FSH, ↑androstenedione, ↓estradiol, ↑estrone)

1. Underlying Neuroendocrine Dysregulation

The hypothalamic-pituitary axis in PCOS is **intrinsically dysregulated**, possibly due to genetic and environmental factors.

This leads to **increased GnRH pulse frequency**, which favors **LH secretion** over FSH; LH:FSH ratio (often >2:1)

2. Androgen Feedback Loop

- LH stimulates theca cells → **ovarian thecal overactivity** → ↑ androgen production (e.g., testosterone, androstenedione) → ↓ FSH (negative feedback loop)
- ↓ FSH → Impaired Granulosa Cell Function (↓ aromatase activity in granulosa cells); Androgens are not efficiently converted to estrogens → **follicular arrest (first meiosis)** and anovulation
- This selective feedback maintains or further increases **GnRH pulse frequency**, reinforcing LH dominance.

3. Peripheral Conversion of Androgens to Estrogens

Androgens → **estrone** via aromatization in adipose tissue

Leads to **chronic low-level estrogen without progesterone** → **endometrial hyperplasia or carcinoma**

4. Insulin Resistance and Hyperinsulinemia

Insulin acts synergistically with LH to stimulate theca cells → **more androgen production**.

Insulin also reduces SHBG (Sex Hormone Binding Globulin) → ↑ **free androgens**, which perpetuate the feedback loop.

5. Absence of Progesterone Feedback

Due to **anovulation**, there is no corpus luteum → **no progesterone** → **menstrual irregularity and unopposed estrogen effects**.

Progesterone normally slows GnRH pulses; its absence allows **faster GnRH pulsate**

Question 10:

A 28-year-old woman comes to the clinic because of 8 months of amenorrhea. She reports hot flashes, night sweats, and vaginal dryness. She has no significant past medical history and takes no medications. Her menarche was at age 13, and her cycles had been regular until the past year. She denies hirsutism or acne. BMI is 22 kg/m². Pelvic exam shows normal external genitalia with reduced vaginal rugae. A pregnancy test is negative. Laboratory studies show:

Serum FSH: markedly elevated

Serum LH: elevated

Serum estradiol: decreased

Serum testosterone: normal

Which of the following is the most likely diagnosis?

- A. Premature ovarian failure (primary ovarian insufficiency)
- B. Polycystic ovarian syndrome (PCOS)
- C. Functional hypothalamic amenorrhea
- D. Asherman syndrome
- E. Hyperprolactinemia

Question 10: Explanation

A. Premature ovarian failure:

- Ovarian failure before age 40, characterized by amenorrhea, hypoestrogenic symptoms (hot flashes, vaginal atrophy), elevated FSH/LH (hypergonadotropic hypogonadism), and low estradiol [ovarian (hypergonadotropic) cause]
- Normal BMI and absence of androgen excess further support the diagnosis.

B. Polycystic ovarian syndrome (PCOS):

- Typically presents with obesity, hirsutism, acne, infertility, and oligomenorrhea/amenorrhea.
- Lab findings include elevated LH/FSH ratio ($>2:1$), increased androgens, and normal/low estrogen
- This patient lacks hyperandrogenic features and has low estrogen with high FSH, not typical of PCOS

C. Functional hypothalamic amenorrhea:

- Seen in patients with low BMI, excessive exercise, or stress.
- Characterized by low or normal FSH, LH, and estrogen due to suppressed GnRH secretion.

D. Asherman syndrome: Outflow-tract/endometrial scarring

- Intrauterine adhesions following curettage or infection cause amenorrhea, but systemic hypoestrogenic symptoms (hot flashes, vaginal atrophy) are absent
- Gonadotropins and estradiol are normal; SH/LH would not be markedly elevated.

E. Hyperprolactinemia:

- Causes galactorrhea, headaches, or visual field defects due to pituitary adenoma
- Labs would show low/normal FSH/LH, low estrogen due to GnRH suppression, not the high FSH/LH seen here

POF vs PCOS

Feature	Premature Ovarian Failure (POF/Primary Ovarian Insufficiency)	Polycystic Ovarian Syndrome (PCOS)
Age	<40 (pathologic)	Usually reproductive age
Symptoms	Amenorrhea, hot flashes, vaginal dryness, infertility	Oligomenorrhea/amenorrhea, infertility, obesity, hirsutism, acne
Hormones	↑ FSH, ↑ LH, ↓ Estrogen, normal testosterone	↑ LH/FSH ratio (>2:1), ↑ Androgens, normal/low Estrogen
Pathophysiology	Loss of ovarian follicles → ovarian failure	Chronic anovulation + hyperandrogenism
Long-term risk	Osteoporosis, CVD	Metabolic syndrome, endometrial hyperplasia/carcinoma
Ovarian morphology	Atrophic ovaries	Enlarged, polycystic ovaries

Question 11:

A 58-year-old woman presents with abdominal bloating, early satiety, and increasing abdominal girth. Physical exam reveals ascites and a palpable adnexal mass. CA-125 is elevated. Imaging shows bilateral complex ovarian masses with peritoneal implants. Biopsy reveals high-grade serous carcinoma.

Which of the following features is most characteristic of this malignancy?

- A.** KRAS mutation and unilateral cystic mass
- B.** TP53 mutation and bilateral solid-cystic masses
- C.** Estrogen-secreting tumor with endometrial hyperplasia
- D.** Mature tissues from all three germ layers
- E.** Mucin-secreting cells with pseudomyxoma peritonei

Ovarian Tumors-Overview

LO: Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord–stromal cells) as the basis for tumor classification.

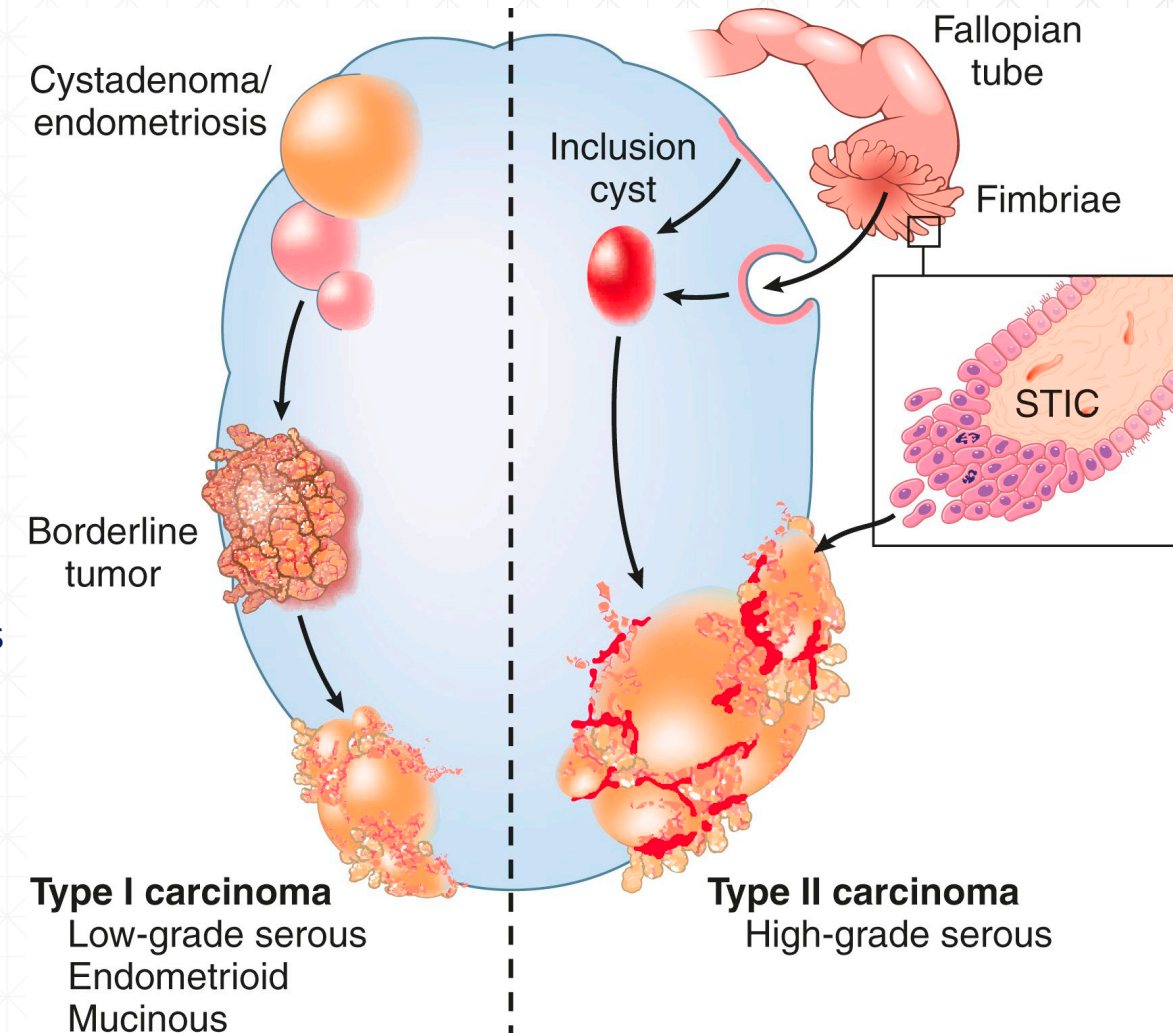
- Ovarian malignancy is the second most common gynecologic malignancy (following endometrial carcinoma)
- Tumors of the ovary arise from any of the three cell types in the normal ovary: (1) multipotent surface/fallopian tube epithelium; (2) pluripotent germ cells; and (3) sex cord–stromal cells.
- The great majority of malignant ovarian tumors are of epithelial cell origin; 90%
- Germ cell and sex cord–stromal cell tumors are less frequent, 10% of malignant tumors of the ovary

Ovarian Tumors-Classification

LO: Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord–stromal cells) as the basis for tumor classification.

Most primary ovarian neoplasms arise from müllerian epithelium:

- Classified based on differentiation and extent of proliferation of the epithelium
- Three major histologic types: serous, mucinous, and endometrioid
- Benign:**
 - 80% of surface epithelial tumors
 - occur in young women between 20 and 45 years of age
 - Frequently cystic and may have an accompanying stromal component
- Borderline:**
 - Occur at older ages
 - Majority behave in a benign manner, some recur, and a few progress to carcinoma
- Malignant tumors:**
 - Between 45 and 65 years of age and may be cystic (cystadenocarcinoma) or solid (carcinoma).
 - Types I and II
- Type I carcinoma:** low-grade tumor; in association with borderline tumors or endometriosis and includes low-grade serous, endometrioid, and mucinous tumors
- Type II carcinoma :** high-grade serous carcinoma that arises from serous tubal intraepithelial carcinoma



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Derivation of various ovarian neoplasms. Type I tumors progress from benign tumors through borderline tumors that may give rise to a low-grade carcinoma. Type II tumors arise from inclusion cysts/fallopian tube epithelium via intraepithelial precu...

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Ovarian Tumors-Risk Factors

LO: Identify risk factors, hereditary cancer syndromes, and molecular mechanisms underlying epithelial, sex cord–stromal, and germ cell ovarian tumors.

Important risk factors:

- increasing age
- early menarche/late menopause
- Nulliparity
- family history
- germline mutations in certain tumor suppressor genes
- **Prolonged use of oral contraceptives reduces the risk due to suppression of ovulation.**
- **Around 5% to 10% of ovarian cancers are familial**, associated with mutations in the *BRCA1* or *BRCA2* tumor suppressor genes (also mutated in a subset of hereditary breast cancer).

Ovarian Tumors: Serous Tumors

LO: Identify risk factors, hereditary cancer syndromes, and molecular mechanisms underlying epithelial, sex cord–stromal, and germ cell ovarian tumors.

Serous Tumors

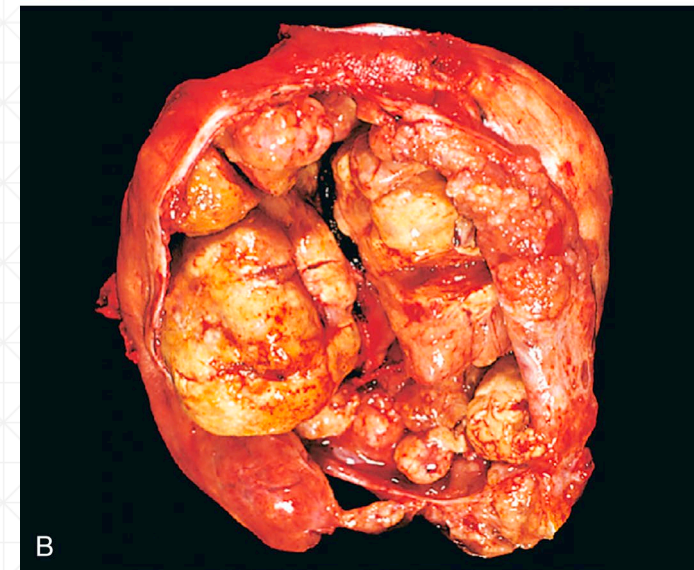
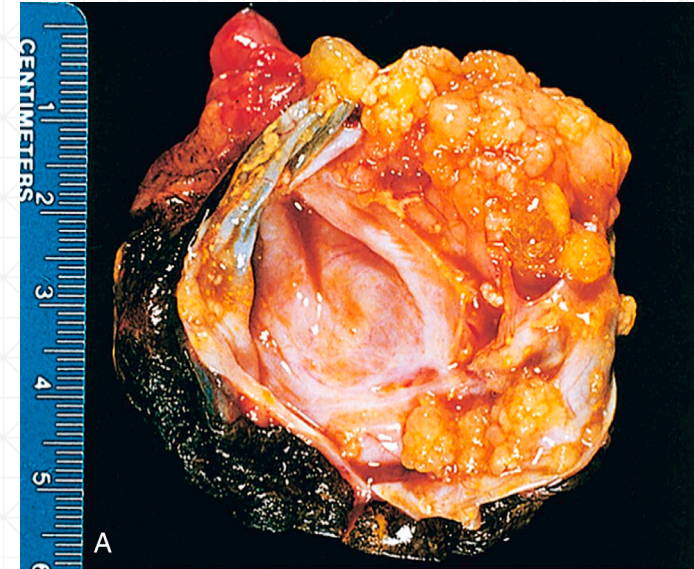
- Most common of the ovarian epithelial tumors
- Benign and borderline tumors are most common between 20 and 45 years of age
- Serous carcinomas generally occur later in life except in familial cases.
- Low-grade tumors arise from benign or borderline lesions and progress slowly in a stepwise manner to become invasive carcinoma.
- Low-grade tumors have mutations in the *KRAS*, *BRAF*, or *ERBB2* oncogenes and wild-type *TP53* alleles
- High-grade serous tumors arise in the fimbriated end of the fallopian tube via serous tubal intraepithelial carcinoma
- High-grade serous tumors have *TP53* mutations; 95% of cases
- Almost all ovarian carcinomas arising in women with *BRCA1* or *BRCA2* mutations are high-grade serous carcinomas with *TP53* mutations

Ovarian Tumors: Serous Tumors- Gross Pathology

LO: Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord–stromal cells) as the basis for tumor classification.

Morphology:

- large, spherical to ovoid, cystic structures up to 30 to 40 cm in diameter.
- About 25% of benign tumors are bilateral.
- In benign tumors, the serosal covering is smooth and glistening.
- By contrast, the surface of adenocarcinoma has nodular irregularities where tumor has invaded the serosa.
- Single cavity, or multiple cavities divided by multiple septa into multiloculated masses
- The cystic spaces are filled with a clear, serous fluid.
- Protruding into the cystic cavities are papillary projections (malignant)



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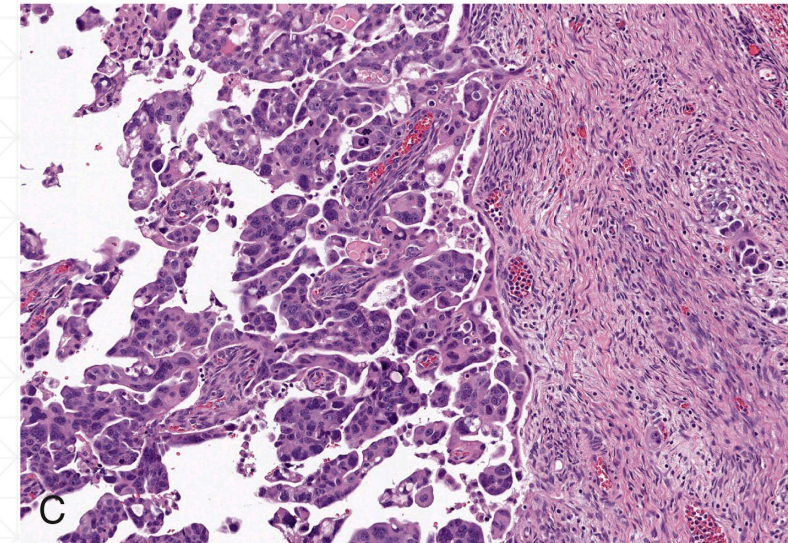
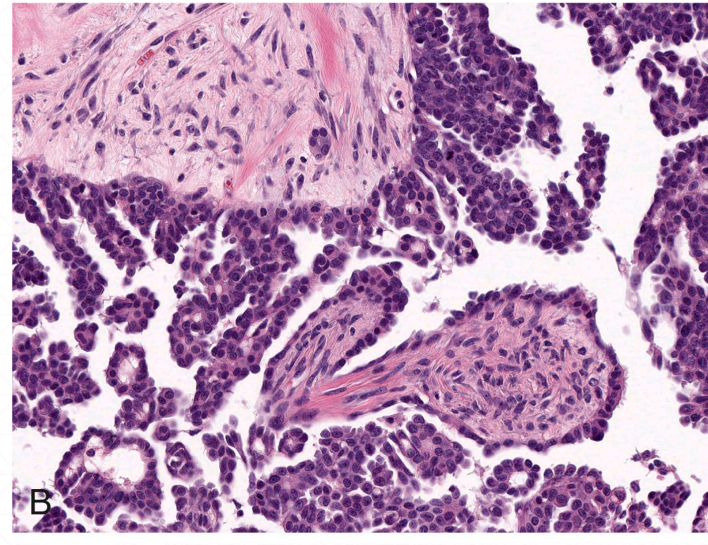
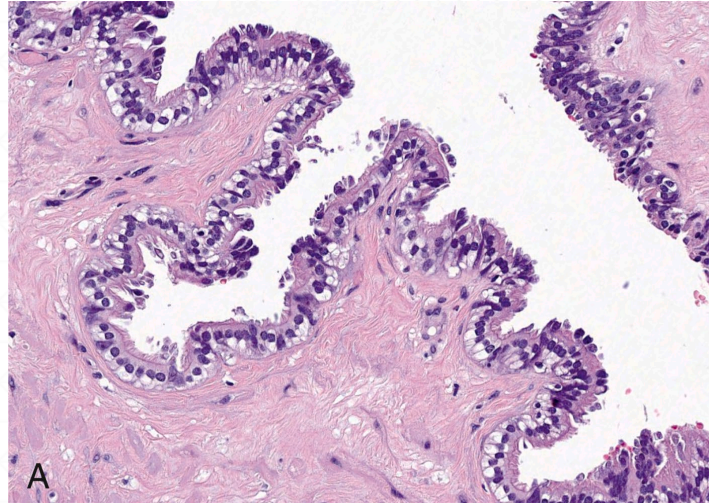
Ovarian serous tumors. (A) Borderline serous tumor opened to display a cyst cavity lined by delicate papillary tumor growths. (B) Cystadenocarcinoma. The cyst is opened to show a large, bulky tumor mass. Courtesy of Dr. Christopher Crum, Brigham a...
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Ovarian Tumors: Serous Tumors- Histopathology

LO: Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord-stromal cells) as the basis for tumor classification.

Microscopic:

- **Benign tumors (serous cystadenoma; A):**
 - A single layer of columnar epithelial cells that line the cyst(s)
 - Cells are ciliated
 - Concentric calcifications (psammoma bodies) are common; not specific
- **High-grade carcinoma; C:**
 - The cells are markedly atypical
 - Papillary formations are complex and multilayered
 - Malignant cells invade the ovarian stroma
- **low-grade carcinoma:** less cytologic atypia
- **Borderline tumors (B):** less cytologic atypia, no stromal invasion; peritoneal implants are “noninvasive”
- Ovarian serous tumors, both low and high grade spread to the peritoneal surfaces and omentum-----commonly associated with ascites



Question 12:

A 45-year-old woman undergoes surgery for a large ovarian mass. The gross picture of the mass is provided. Histology shows epithelial stratification, nuclear atypia, and stromal invasion. Genetic testing reveals a KRAS mutation.

Which of the following best describes the tumor type?

- A.** High-grade serous carcinoma
- B.** Mucinous cystadenocarcinoma
- C.** Endometrioid carcinoma
- D.** Mature teratoma
- E.** Clear cell carcinoma

Ovarian Tumors: Mucinous Tumors

LO: Identify risk factors, hereditary cancer syndromes, and molecular mechanisms underlying epithelial, sex cord–stromal, and germ cell ovarian tumors.

Mucinous Tumors

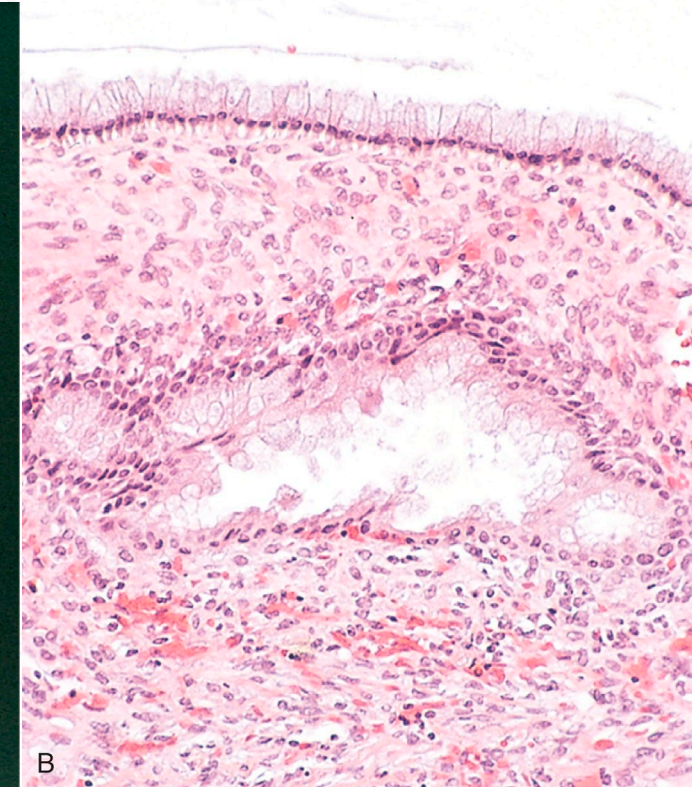
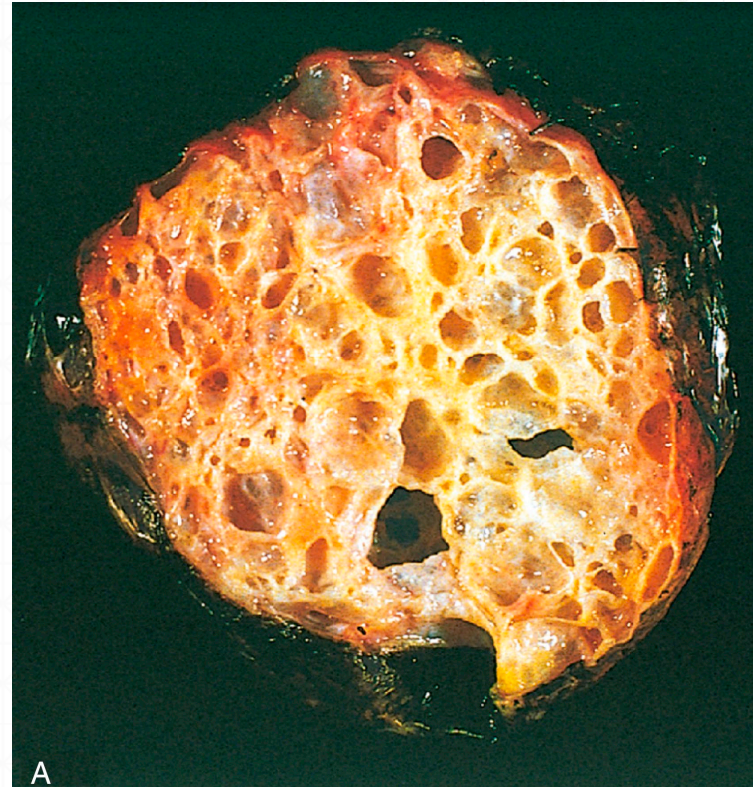
- Two features distinguish them from serous tumors:
 - Mucin-secreting cells
 - Mucinous tumors are less likely to be malignant
 - Overall, only 10% of mucinous tumors are malignant; another 10% are borderline, and 80% are benign.
 - Occur in middle adult life and are rare before puberty and after menopause
 - **Mutation of the *KRAS* proto-oncogene is a consistent genetic alteration in all mucinous tumors of the ovary.**

Ovarian Tumors: Mucinous Tumors-Morphology

LO: Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord–stromal cells) as the basis for tumor classification.

Morphology:

- Less bilateral; DDX **Krukenberg tumor**
- Well-circumscribed multicystic masses with smooth inner surface, and abundant mucinous content (yellow tan)
- Mucin-producing epithelial cells line the cysts; a single layer of columnar epithelium with basal nuclei and apical mucin; no stratification, atypia, or stromal invasion
- Malignant tumors are characterized by solid areas of growth, stratification of lining cells, cytologic atypia, and stromal invasion
- Implantation of mucinous tumor cells in the peritoneum---copious amounts of mucin is called **pseudomyxoma peritonei**; caused by metastases from the gastrointestinal tract, usually the appendix



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Ovarian mucinous cystadenoma. (A) Mucinous cystadenoma with multicystic appearance and delicate septa. Note the presence of glistening mucin within the cysts. (B) Columnar cell lining of mucinous cystadenoma. Gross and histological images showing ...

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Ovarian Tumors: Endometrioid and Clear Cell

LO: Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord–stromal cells) as the basis for tumor classification. Identify risk factors, hereditary cancer syndromes, and molecular mechanisms underlying epithelial, sex cord–stromal, and germ cell ovarian tumors.

Endometrioid Tumors

- solid or cystic
- In association with endometriosis
- Microscopic examination: tubular glands, similar to those of the endometrium, within the lining of cystic spaces
- Usually malignant; benign and borderline forms exist
- They are bilateral in about 20% of cases, and 15% to 20% of women with these ovarian tumors have a concomitant endometrial carcinoma.
- Similar to endometrioid-type carcinoma of the endometrium, endometrioid carcinomas of the ovary frequently have mutations in the *PTEN* tumor suppressor gene and other genes that act to upregulate PI3K-AKT signaling

Clear Cell Carcinoma

- A rare subtype of epithelial ovarian tumors
- Benign and borderline are rare
- Clear cell carcinomas are composed of large epithelial cells with abundant clear cytoplasm (similar to gestational endometrium)
- In association with endometriosis or endometrioid carcinoma of the ovary
- Most commonly mutated genes (*PIK3CA*, *ARID1A*, *KRAS*, *PTEN*, and *TP53*) are shared with endometrioid carcinoma

Question 13:

A 20-year-old woman presents with acute pelvic pain. Imaging reveals a 6 cm ovarian mass with calcifications and fat. Surgical removal reveals a cyst filled with hair and sebaceous material. Histology shows skin, cartilage, and respiratory epithelium.

Which of the following best describes this tumor?

- A.** Immature teratoma
- B.** Dysgerminoma
- C.** Mature cystic teratoma
- D.** Sertoli-Leydig cell tumor
- E.** Endometrioid carcinoma

Ovarian Teratomas

LO: Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord–stromal cells) as the basis for tumor classification. Identify risk factors, hereditary cancer syndromes, and molecular mechanisms underlying epithelial, sex cord–stromal, and germ cell ovarian tumors.

- **Teratomas account for 15–20% of ovarian tumors; >90% are benign, most commonly found in young women.**
- They are classified into:
 - Mature (benign);
 - Immature (malignant)
 - Monodermal or highly specialized

Mature (Benign) Teratomas (dermoid cysts):

- Mature tissues from all three germ layers:
 - Ectoderm: skin, hair, neural tissue
 - Mesoderm: bone, cartilage, muscle
 - Endoderm: respiratory or gastrointestinal epithelium

Gross features:

Cystic structure filled with sebaceous material and hair

May contain teeth, bone, cartilage

Usually unilateral (90%), more common on the right side

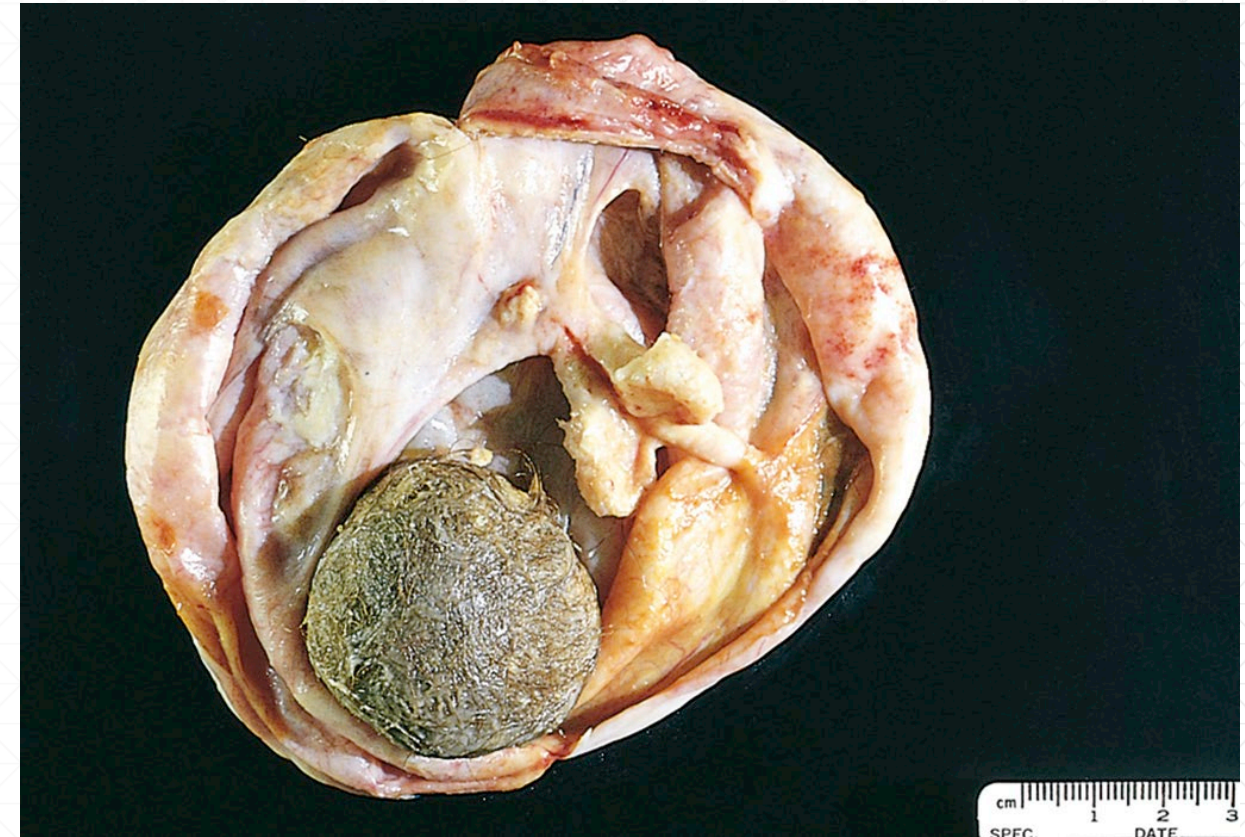
Rarely exceeds 10 cm

Complications:

Ovarian torsion (10–15% of cases)

Malignant transformation (~1%, usually to squamous cell carcinoma)

Paraneoplastic limbic encephalitis (rare, associated with neural tissue)



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Mature cystic teratoma (dermoid cyst) of the ovary. A ball of hair (bottom) and a mixture of tissues are evident. Courtesy of Dr. Christopher Crum, Brigham and Women's Hospital, Boston, Massachusetts. Gross specimen image of ovarian mature cystic ...

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Ovarian Teratomas: Immature (Malignant Teratomas)

LO: Describe the embryologic/histologic components of the ovary (surface epithelium, germ cells, sex cord–stromal cells) as the basis for tumor classification. Identify risk factors, hereditary cancer syndromes, and molecular mechanisms underlying epithelial, sex cord–stromal, and germ cell ovarian tumors.

Immature (Malignant) Teratomas

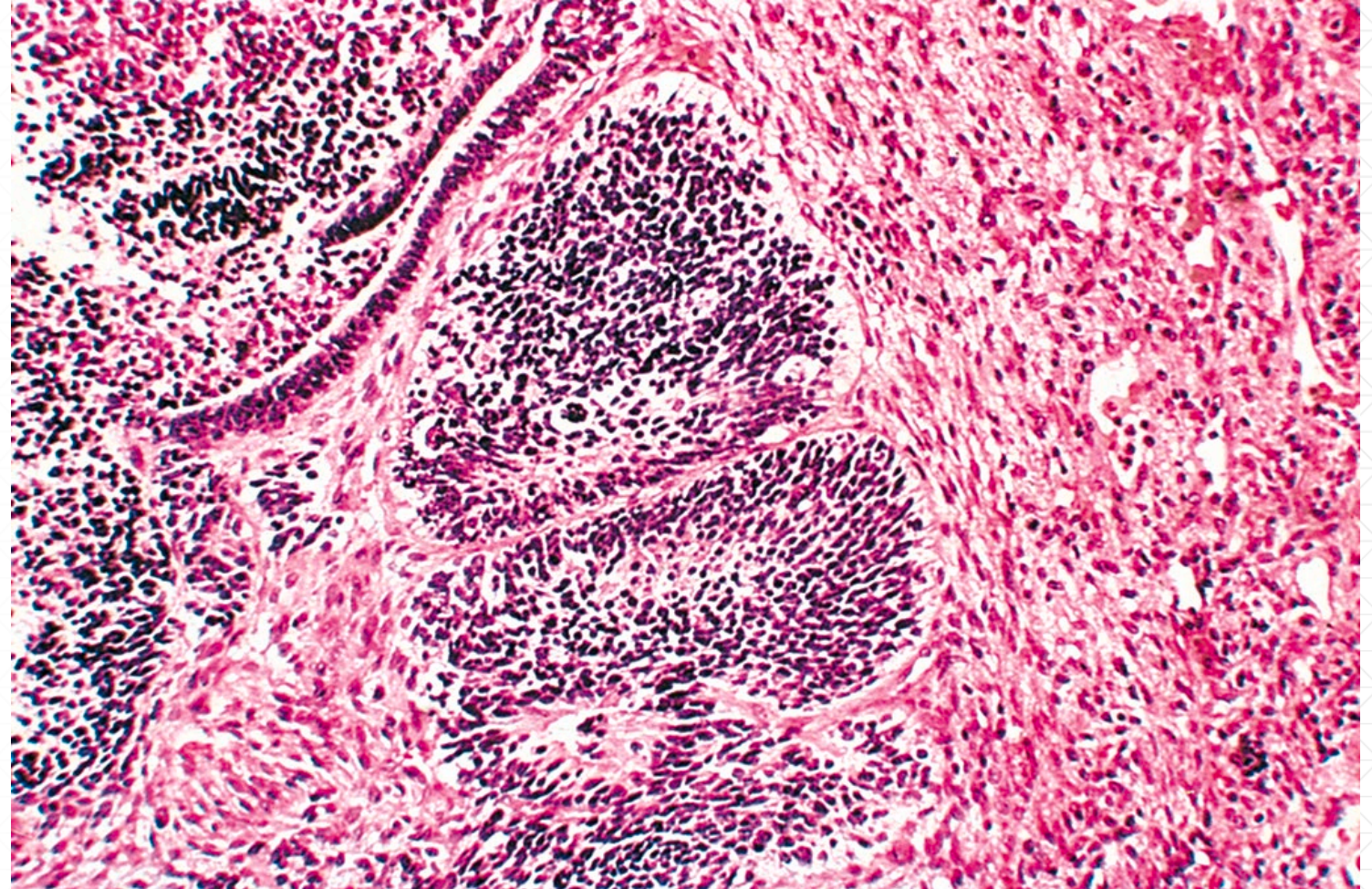
- Found in prepubertal adolescents and young women (mean age: 18)
- Solid and bulky on gross examination
- Composed of immature fetal-like tissues, especially:
 - Neuroepithelium
 - Cartilage
 - Bone

Histological features:

- Primitive neuroepithelial tubules
- Hyperchromatic, densely packed cells
- Rosette-like structures

Prognosis:

Risk of metastasis correlates with the amount of immature neuroepithelium



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Immature teratoma of the ovary illustrating primitive neuroepithelium. Histological section shows immature ovarian teratoma with primitive neuroepithelial tubules composed of hyperchromatic, densely packed cells. Histological section demonstrates ...

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Question 14:

A 3-year-old girl is evaluated for early breast development and intermittent vaginal bleeding. She has no pubic or axillary hair. Physical examination reveals Tanner stage II breast development. Pelvic ultrasound reveals a unilateral ovarian mass.

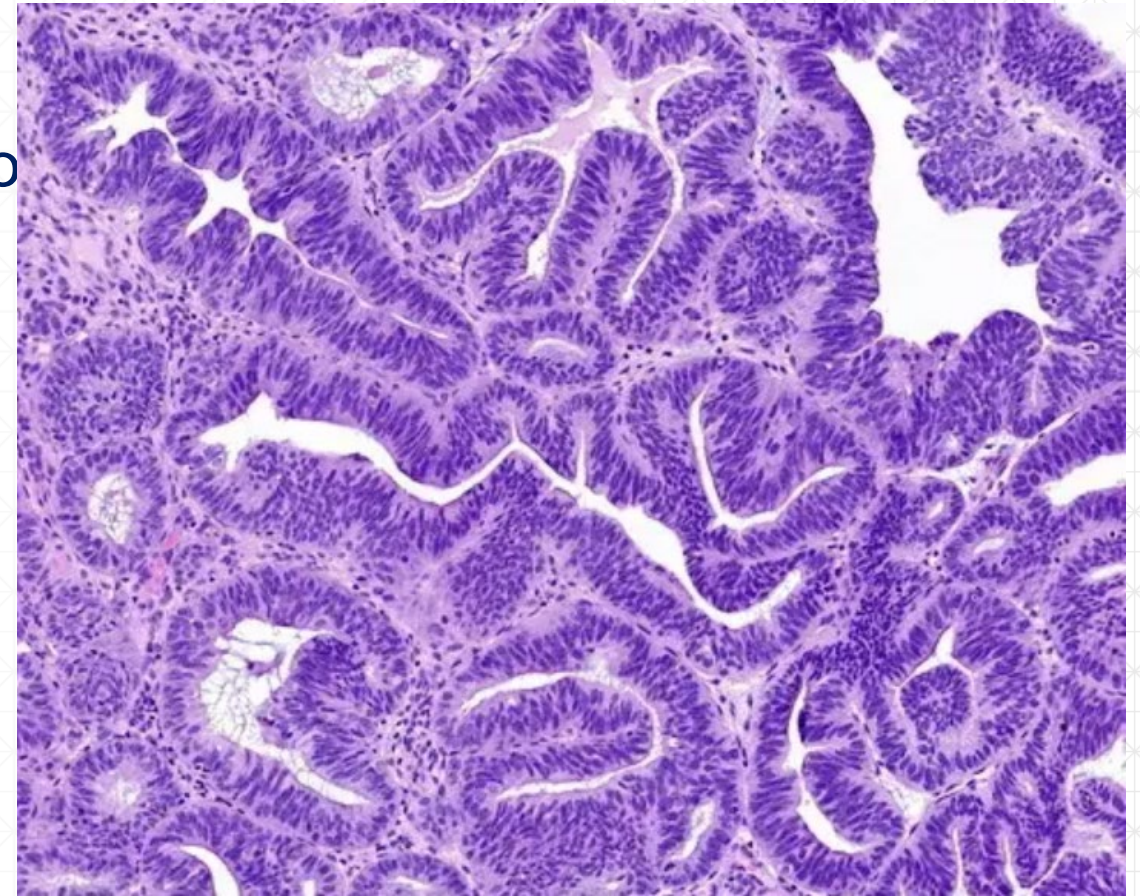
Choice	Estradiol	FSH	LH
A	↑	↓	↓
B	↑	↑	↑
C	↓	↑	↑
D	↑	↓	↑
E	↑	Normal	Normal

Question 15:

A 55-year-old woman presents with vaginal bleeding. Pelvic ultrasound reveals a unilateral ovarian mass. A photomicrograph of an endometrial biopsy is provided. Histologic examination of the ovarian mass reveals Call-Exner bodies and cells with coffee-bean nuclei.

Which of the following is the most likely diagnosis?

- A. Granulosa cell tumor
- B. Sertoli-Leydig cell tumor
- C. Serous cystadenocarcinoma
- D. Immature teratoma
- E. Dysgerminoma



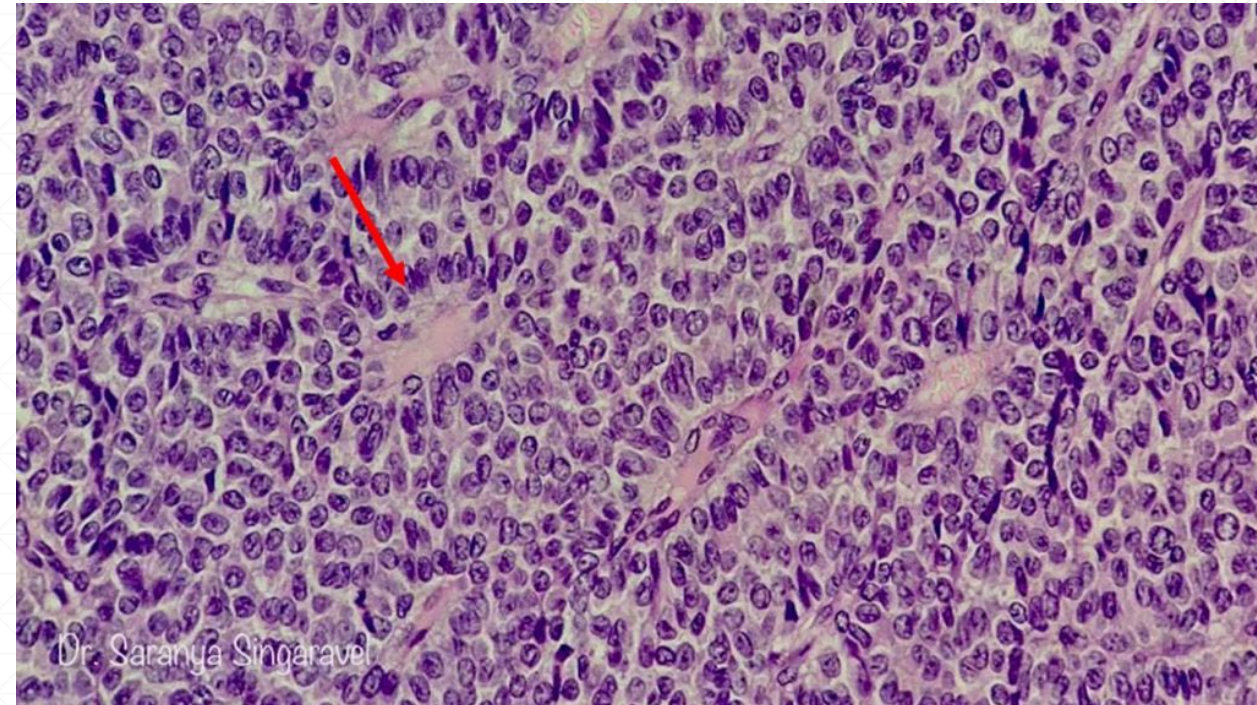
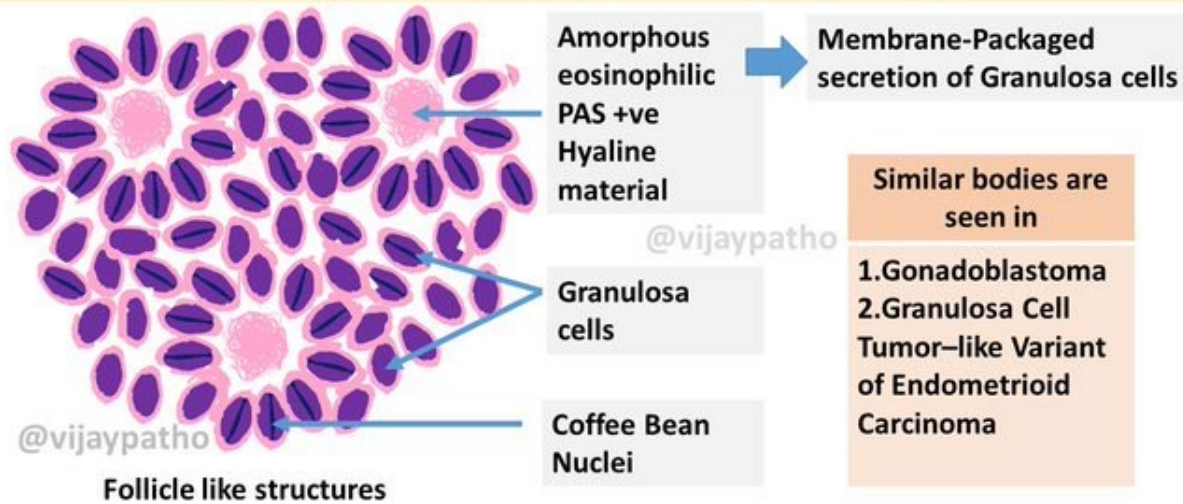
Granulosa Cell Tumor

- Granulosa cell tumor (GCT) of the ovary belongs to the category of **sex cord-stromal tumors** that show differentiation towards **granulosa cells** of the **ovarian follicle**. There are **two subtypes** - **adult GCT** and **juvenile GCT**.
- Adult GCT**, mostly seen in **peri- and postmenopausal women**, accounts for **95%** of all GCTs and makes up **1-2% of all ovarian tumors**--Postmenopausal bleeding, breast tenderness, pelvic mass
- Juvenile GCT**, which occurs in **children and younger women (<30 yrs)**, makes up the remainder **5%** of all GCTs--precocious puberty, breast development and vaginal bleeding in toddlers

Call- Exner Bodies

Emma Louise Call- American obstetrician
Sigmund Exner -Austrian physiologist and psychologist.

Pathognomonic of Adult Granulosa cell tumors



Small, bland, cuboidal to polygonal cells arranged in various patterns. **Call-Exner bodies** (Red Arrow): small follicle-like structures, filled with eosinophilic material. The cells may show **nuclear grooves** (coffee bean nuclei).

Thank You

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